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Binding machine

Abstract

A reinforcing bar binding machine (1A) includes a magazine (2A) in which a wire (W) is housed, a curl guide unit (5A) that winds the wire (W) around a reinforcing bar(S), a wire feeding unit (3A) that feeds the wire (W), and a binding unit (7A) that twists crossing portions between one end and the other end of the wire (W) wound around the reinforcing bar(S). The wire feeding unit (3A) is configured such that a second displacement member (36) for displacing a second feed gear (30R) is located behind the second feed gear (30R) and provided between the second feed gear (30R) and a handle portion (11A).

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application of U.S. application Ser. No. 15/746,128 filed Jan. 19, 2018, which is the 35 U.S.C. 371 National Phase Entry Application of PCT/JP2016/071425, filed Jul. 21, 2016, which claims priority to Japanese Patent Application No. 2015-145285, filed Jul. 22, 2015 and Japanese Patent Application No. 2016-136069, filed Jul. 8, 2016, the disclosures of which are incorporated herein in their entirety by reference and priority is claimed to each.

TECHNICAL FIELD

(1) The present invention relates to a binding machine for binding a binding object such as reinforcing bars with a wire.

BACKGROUND ART

(2) In the related art, there has been suggested a binding machine called a reinforcing bar binding machine which winds a wire around two or more reinforcing bars and twists the wound wire to bind the two or more reinforcing bars.

(3) A reinforcing bar binding machine according to the related art has a configuration in which wires are fed and wound around a reinforcing bar, and then are twisted and bound (for example, see Patent Literature 1). In order to reduce the amount of wire used for such a reinforcing bar binding machine, a reinforcing bar binding machine has been proposed in which wires are fed in a forward direction and wound around the reinforcing bar, and then the wires are fed (pulled back) in a backward direction and wound around the reinforcing bar in close contact with the reinforcing bar (for example, see Patent Literature 2).

(4) Both of the reinforcing bar binding machines have a configuration in which a pair of feeding members sandwich a wire therebetween and then feed the wire. The pair of feeding members are close to each other due to the need of the sandwiching of the wire during feeding of the wire, but are separated from each other during removal of the wire from the feeding member or loading of

the wire on the feeding member. For this reason, at least one of the pair of feeding members includes a displacement member for contacting and separating to and from the other feeding member. The displacement member is pressed by a spring or the like so that one feeding member maintains a state of being in contact with or separated from the other feeding member. In addition, the reinforcing bar binding machine includes an operation member that operates the displacement member.

CITATION LIST

Patent Literature

(5) [Patent Literature 1]: Japanese Unexamined Utility Model Application Publication No. H7-34110 [Patent Literature 2]: Japanese Patent No. 3680804

SUMMARY

Technical Problem

(6) In the reinforcing bar binding machine according to the related art, the displacement member for displacing the pair of feeding members and the operation member for operating the displacement member are provided between a feeding member and a magazine in which a reel wound with wires is housed. Thus, a space is required between the feeding member and the magazine to provide the displacement member and the operation member, and thus the magazine is not freely disposed. Further, the displacement member and the operation member are present in a feed path of the wire passing through the feeding member, and this causes hindrance of wire loading.

(7) The present invention has been made to solve such problems, and an object thereof is to provide a binding machine which reduces constraints on arrangement of a housing unit in which a wire is housed.

Solution to Problem

(8) In order to solve the above-described problems, the present invention provides a binding machine includes: a housing in which a wire is housed in a drawable manner; a feeding unit that includes a pair of feeding members between which the wire is sandwiched and fed, and that is capable of winding the wire around a binding object; a binding unit that twists the wire wound by the feeding unit; and a displacement unit that displaces the pair of feeding members in a direction coming in contact with and separating from each other, wherein the displacement unit is provided in a direction intersecting to a feeding direction of the wire with respect to the feeding member.

Advantageous Effects of the Invention

(9) According to the present invention, the displacement unit of the feeding member is not present between the housing and the feeding member, and thus the degree of freedom in arrangement of the housing is improved. This makes it possible to downsize the device. In addition, the feed path of the wire passing through the pair of feeding member does not have a configuration for hindering the wire loading, and thus the wire loading can easily be performed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view of an example of an overall configuration of a reinforcing bar binding machine of the present embodiment as viewed from the side.

(2) FIG. 2 is a front view illustrating an example of the overall configuration of the reinforcing bar binding machine of the present embodiment as viewed from the front.

(3) FIG. 3 is a view illustrating an example of a feed gear according to the present embodiment.

(4) FIG. 4A is a view illustrating an example of a displacement unit of the present embodiment.

(5) FIG. 4B is a view illustrating an example of a displacement unit of the present embodiment.

(6) FIG. 4C is a view illustrating an example of a displacement unit of the present embodiment.

- (7) FIG. 4D is a view illustrating an example of a displacement unit of the present embodiment.
- (8) FIG. 5A is a view illustrating an example of a parallel guide of the present embodiment.
- (9) FIG. 5B is a view illustrating an example of a parallel guide of the present embodiment.
- (10) FIG. 5C is a view illustrating an example of a parallel guide of the present embodiment.
- (11) FIG. 5D is a view illustrating an example of parallel wires.
- (12) FIG. 5E is a view illustrating an example of intersecting twisted wires.
- (13) FIG. 6 is a view illustrating an example of a guide groove of the present embodiment.
- (14) FIG. 7 is a view illustrating an example of a second guide unit of the present embodiment.
- (15) FIG. 8A is a view illustrating an example of a second guide unit of the present embodiment.
- (16) FIG. 8B is a view illustrating an example of a second guide unit of the present embodiment.
- (17) FIG. 9A is a view illustrating an example of a second guide unit of the present embodiment.
- (18) FIG. 9B is a view illustrating an example of a second guide unit of the present embodiment.
- (19) FIG. 10A is a view illustrating main parts of a gripping unit according to the present embodiment.
- (20) FIG. 10B is a view illustrating main parts of a gripping unit according to the present embodiment.
- (21) FIG. 11 is an external view illustrating an example of the reinforcing bar binding machine of the present embodiment.
- (22) FIG. 12 is an explanatory view of an operation of the reinforcing bar binding machine of the present embodiment.
- (23) FIG. 13 is an explanatory view of an operation of the reinforcing bar binding machine of the present embodiment.
- (24) FIG. 14 is an explanatory view of an operation of the reinforcing bar binding machine of the present embodiment.
- (25) FIG. 15 is an explanatory view of an operation of the reinforcing bar binding machine of the present embodiment.
- (26) FIG. 16 is an explanatory view of an operation of the reinforcing bar binding machine of the present embodiment.
- (27) FIG. 17 is an explanatory view of an operation of the reinforcing bar binding machine of the present embodiment.
- (28) FIG. 18 is an explanatory view of an operation of the reinforcing bar binding machine of the present embodiment.
- (29) FIG. 19 is an explanatory view of an operation of the reinforcing bar binding machine of the present embodiment.
- (30) FIG. 20A is an explanatory view of an operation of winding a wire around a reinforcing bar.
- (31) FIG. 20B is an explanatory view of an operation of winding a wire around a reinforcing bar.
- (32) FIG. 20C is an explanatory view of an operation of winding a wire around a reinforcing bar.
- (33) FIG. 21A is an explanatory view of an operation of forming a loop with a wire by a curl guide unit.
- (34) FIG. 21B is an explanatory view of an operation for forming a loop with a wire by a curl guide unit.
- (35) FIG. 22A is an explanatory view of an operation of bending a wire.
- (36) FIG. 22B is an explanatory view of an operation of bending the wire.
- (37) FIG. 22C is an explanatory view of an operation of bending the wire.
- (38) FIG. 23A is an operational effect example of the reinforcing bar binding machine of the present embodiment.
- (39) FIG. 23B is an operational effect example of the reinforcing bar binding machine of the present embodiment.
- (40) FIG. 24A is an operational effect example of the reinforcing bar binding machine of the present embodiment.

(41) FIG. 24B is an example of the operation and problem of the reinforcing bar binding machine according to the related art.

(42) FIG. 25A is an operational effect example of the reinforcing bar binding machine of the present embodiment.

(43) FIG. 25B is an example of the operation and problem of the reinforcing bar binding machine according to the related art.

(44) FIG. 26A is an operational effect example of the reinforcing bar binding machine of the present embodiment.

(45) FIG. 26B is an example of the operation and problem of the reinforcing bar binding machine according to the related art.

(46) FIG. 27A is an operational effect example of the reinforcing bar binding machine of the present embodiment.

(47) FIG. 27B is an operational effect example of the reinforcing bar binding machine of the present embodiment.

(48) FIG. 27C is an example of the operation and problem of the reinforcing bar binding machine according to the related art.

(49) FIG. 27D is an example of the operation and problem of the reinforcing bar binding machine according to the related art.

(50) FIG. 28A is an operational effect example of the reinforcing bar binding machine of the present embodiment.

(51) FIG. 28B is an example of the operation and problem of the reinforcing bar binding machine according to the related art.

(52) FIG. 29A is a view illustrating a modified example of the second guide unit of the present embodiment.

(53) FIG. 29B is a view illustrating a modified example of the second guide unit of the present embodiment.

(54) FIG. 30A is a view illustrating a modified example of the parallel guide of the present embodiment.

(55) FIG. 30B is a view illustrating a modified example of the parallel guide of the present embodiment.

(56) FIG. 30C is a view illustrating a modified example of the parallel guide of the present embodiment.

(57) FIG. 30D is a view illustrating a modified example of the parallel guide of the present embodiment.

(58) FIG. 30E is a view illustrating a modified example of the parallel guide of the present embodiment.

(59) FIG. 31 is a view illustrating a modified example of the guide groove of the present embodiment.

(60) FIG. 32 is a view illustrating an example of a displacement unit according to another embodiment.

(61) FIG. 33 is a view illustrating an example of the displacement unit according to another embodiment.

(62) FIG. 34 is a view illustrating an example of the displacement unit according to another embodiment.

(63) FIG. 35 is an external view illustrating an example of a reinforcing bar binding machine according to another embodiment.

(64) FIG. 36 is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment FIG. 37 is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment.

(65) FIG. 38 is an explanatory view illustrating an example of an operation of a displacement unit

according to another embodiment.

(66) FIG. **39** is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment.

(67) FIG. **40** is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment.

(68) FIG. **41** is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment. FIG. **42** is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment.

(69) FIG. **43** is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment.

(70) FIG. **44** is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment.

(71) FIG. **45** is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment.

(72) FIG. **46** is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment.

(73) FIG. **47** is an explanatory view illustrating an example of an operation of a displacement unit according to another embodiment.

(74) FIG. **48** is an external view illustrating an example of a reinforcing bar binding machine according to another embodiment.

DETAILED DESCRIPTION

(75) Hereinafter, an example of a reinforcing bar binding machine as an embodiment of a binding machine of the present invention will be described with reference to the drawings.

(76) <Example of Configuration of Reinforcing Bar Binding Machine of this Embodiment>

(77) FIG. **1** is a view of an example of the overall configuration of a reinforcing bar binding machine according to the present embodiment as seen from a side, and FIG. **2** is a view illustrating an example of the overall configuration of the reinforcing bar binding machine of the present embodiment as seen from a front. Here, FIG. **2** schematically illustrates the internal configuration of the line A-A in FIG. **1**.

(78) The reinforcing bar binding machine **1A** of the present embodiment binds the reinforcing bar **S**, which is a binding object, by using two or more wires **W** having a diameter smaller compared to a conventional wire having a large diameter. In the reinforcing bar binding machine **1A**, as will be described later, by the operation of winding the wire **W** around the reinforcing bar **S**, the operation of winding the wire **W** wound around the reinforcing bar **S** in close contact with the reinforcing bar **S**, and the operation of twisting the wire wound around the reinforcing bar **S**, the reinforcing bar **S** is bound with the wire **W**. In the reinforcing bar binding machine **1A**, since the wire **W** is bent in any of the operations described above, by using the wire **W** having a smaller diameter than the conventional wire, the wire is wound on the reinforcing bar **S** with less force, and it is possible to twist the wire **W** with less force. Further, by using two or more wires, it is possible to secure the binding strength of the reinforcing bar **S** by the wire **W**. In addition, by arranging two or more wires **W** to be fed in parallel, the time required for winding the wire **W** can be shortened compared with the operation of winding the reinforcing bar twice or more with one wire. It should also be noted that winding the wire **W** around the reinforcing bar **S** and winding the wire **W** wound around the reinforcing bar **S** in close contact with the reinforcing bar **S** is collectively referred to as winding the wire **W**. The wire **W** may be wound on a binding object other than the reinforcing bar **S**. Here, as the wire **W**, a single wire or a twisted wire made of a metal that can be plastically deformed is used.

(79) The reinforcing bar binding machine **1A** includes a magazine **2A** that is a housing unit that houses the wire **W**, a wire feeding unit **3A** that feeds the wire **W** housed in the magazine **2A**, a parallel guide **4A** for arranging the wires **W** fed to the wire feeding unit **3A** and the wires **W** fed out

from the wire feeding unit 3A in parallel. The reinforcing bar binding machine 1A further includes a curl guide unit 5A that winds the wires W fed out in parallel around the reinforcing bar S, and a cutting unit 6A that cuts the wire W wound around the reinforcing bar S. Further, the reinforcing bar binding machine 1A includes a binding unit 7A that grips and twists the wire W wound around the reinforcing bar S.

(80) The magazine 2A is an example of a housing unit. In the embodiment, a reel 20, having two long wires W wound thereon in a drawable manner, is detachably housed in the magazine. The reel 20 is provided with a tubular hub portion 20a that can wind the wires W and a pair of flanges 20b that are provided at opposite end sides of the hub portion 20a in an axial direction. The flanges 20b have a larger diameter than the hub portion 20a, and protrudes beyond the opposite end sides of the hub portion 20a in the axial direction. Two or more wires W, in this example, two wires W are wound around the hub portion 20a. In the reinforcing bar binding machine 1A, while the reel 20 housed in the magazine 2A rotates, the two wires W are fed out from the reel 20 through the operation of feeding the two wires W by the wire feeding unit 3A and the operation of feeding the two wires W manually. At this time, the two wires W are wound around the hub portion 20a so that the two wires W are fed out without being twisted.

(81) The wire feeding unit 3A is an example of a wire feeding unit constituting the feeding unit and includes a first gear 30L in the form of a spur gear which feeds the wire W by a rotation operation as a pair of feeding members for feeding the parallel wires W, and a second feed gear 30R also having a spur gear shape for sandwiching the wire W with the first feed gear 30L. Although the details of the first feed gear 30L and the second feed gear 30R will be described later, the first feed gear 30L and the second feed gear 30R have a spur gear shape in which teeth are formed on the outer peripheral surface of a disk-like member. However, the first feed gear 30L and the second feed gear 30R are meshed with each other, and the driving force is transmitted from one feed gear to the other feed gear, so that the two wires W can be appropriately fed, it is not necessarily limited to a spur gear shape.

(82) The first feed gear 30L and the second feed gear 30R are an example of a rotary feeding member, and are each formed of a disc-shaped member. In the wire feeding unit 3A, the first feed gear 30L and the second feed gear 30R are provided so as to sandwich the feed path of the wire W, so that the outer peripheral surfaces of the first feed gear 30L and the second feed gear 30R face each other. The first feed gear 30L and the second feed gear 30R sandwich the two parallel wires W between a portion opposing to the outer peripheral surface. The first feed gear 30L and the second feed gear 30R feed two wires W along the extending direction of the wire W in a state where the two wires W are arranged in parallel with each other.

(83) FIG. 3 is a view illustrating an example of the feed gear of this embodiment. Here, FIG. 3 is a sectional view taken along the line B-B of FIG. 2. The first feed gear 30L includes a tooth portion 31L on its outer peripheral surface. The second feed gear 30R includes a tooth portion 31R on its outer peripheral surface.

(84) The first feed gear 30L and the second feed gear 30R are arranged in parallel with each other so that the teeth portions 31L and 31R face each other. In other words, the first feed gear 30L and the second feed gear 30R are arranged in parallel in a direction along the axial direction Ru1 of the loop Ru formed by the wire W wound by the curl guide unit 5A, that is, along the axial direction of the virtual circle in which the loop Ru formed by the wire W is regarded as a circle. In the following description, the axial direction Ru1 of the loop Ru formed by the wire W wound by the curl guide unit 5A is also referred to as the axial direction Ru1 of the loop of wire W.

(85) The first feed gear 30L includes a first feed groove 32L on its outer peripheral surface. The second feed gear 30R includes a second feed groove 32R on its outer peripheral surface. The first feed gear 30L and the second feed gear 30R are arranged such that the first feed groove 32L and the second feed groove 32R face each other.

(86) The first feed groove 32L is formed in a V-groove shape on the outer peripheral surface of the

first feed gear **30L** along the rotation direction of the first feed gear **30L**. The first feed groove **32L** has a first inclined surface **32La** and a second inclined surface **32Lb** forming a V-shaped groove. The first feed groove **32L** has a V-shaped cross section so that the first inclined surface **32La** and the second inclined surface **32Lb** face each other at a predetermined angle. When the wires **W** are held between the first feed gear **30L** and the second feed gear **30R** in parallel, the first feed groove **32L** is configured such that one wire among the outermost wires of the wires **W** arranged in parallel, in this example, a part of the outer peripheral surface of one wire **W1** of the two wires **W** arranged in parallel is in contact with the first inclined surface **32La** and the second inclined surface **32Lb**.

(87) The second feed groove **32R** is formed in a V-groove shape on the outer peripheral surface of the second feed gear **30R** along the rotation direction of the second feed gear **30R**. The second feed groove **32R** has a first inclined surface **32Ra** and a second inclined surface **32Rb** that form a V-shaped groove. Similarly to the first feed groove **32L**, the second feed groove **32R** has a V-shaped cross-sectional shape, and the first inclined surface **32Ra** and the second inclined surface **32Rb** face each other at a predetermined angle. When the wire **W** is held between the first feed gear **30L** and the second feed gear **30R** in parallel, the second feed groove **32R** is configured such that, the other wire among the outermost wires of the wires **W** arranged in parallel, in this example, a part of the outer peripheral surface of the other wire **W2** of the two wires **W** arranged in parallel is in contact with the first inclined surface **32Ra** and the second inclined surface **32Rb**.

(88) When the wire **W** is pinched between the first feed gear **30L** and the second feed gear **30R**, the first feed groove **32L** is configured with a depth and an angle (between the first inclined surface **32La** and the second inclined surface **32Lb**) such that a part, on the side facing the second feed gear **30R**, of one wire **W1** in contact with the first inclined surface **32La** and the second inclined surface **32Lb** protrudes from the tooth bottom circle **31La** of the first feed gear **30**.

(89) When the wire **W** is pinched between the first feed gear **30L** and the second feed gear **30R**, the second feed groove **32R** is configured with a depth and an angle (between the first inclined surface **32Ra** and the second inclined surface **32Rb**) such that a part, on the side facing the first feed gear **30L**, of the other wire **W2** in contact with the first inclined surface **32Ra** and the second inclined surface **32Rb** protrudes from the tooth bottom circle **31Ra** of the second feed gear **30R**.

(90) As a result, the two wires **W** pinched between the first feed gear **30L** and the second feed gear **30R** are arranged such that one wire **W1** is pressed against the first inclined surface **32La** and the second inclined surface **32Lb** of the first feed groove **32L**, and the other wire **W2** is pressed against the first inclined surface **32Ra** and the second inclined surface **32Rb** of the second feeding groove **32R**. Then, one wire **W1** and the other wire **W2** are pressed against each other. Therefore, by rotation of the first feed gear **30L** and the second feed gear **30R**, the two wires **W** (one wire **W1** and the other wire **W2**) are simultaneously fed between the first feed gear **30L** and the second feed gear **30R** while being in contact with each other. In this example, the first feed groove **32L** and the second feed groove **32R** have a V-shaped cross-sectional shape, but it is not necessarily limited to the V-groove shape, and it may be, for example, a trapezoidal shape or an arcuate shape. Further, in order to transmit the rotation of the first feed gear **30L** to the second feed gear **30R**, between the first feed gear **30L** and the second feed gear **30R**, a transmission mechanism including an even number of gears or the like for rotating the first feed gear **30L** and the second feed gear **30R** in opposite directions to each other may be provided.

(91) The wire feeding unit **3A** includes a driving unit **33** for driving the first feed gear **30L** and a displacement unit **34** for pressing and separating the second feed gear **30R** against the first feed gear **30L**.

(92) The driving unit **33** includes a feed motor **33a** for driving the first feed gear **30L** and a transmission mechanism **33b** including a combination of a gear and the like for transmitting the driving force of the feed motor **33a** to the first feed gear **30L**.

(93) In the first feed gear **30L**, the rotation operation of the feed motor **33a** is transmitted via the

transmission mechanism **33b** and the first feed gear rotates. In the second feed gear **30R**, the rotation operation of the first feed gear **30L** is transmitted to the tooth portion **31R** via the tooth portion **31L** and the second feed gear rotates in accordance with the first feed gear **30L**.

(94) As a result, by the rotation of the first feed gear **30L** and the second feed gear **30R**, due to the frictional force generated between the first feed gear **30L** and the one wire **W1**, the friction force generated between the second feed gear **30R** and the other wire **W2**, and the frictional force generated between the one wire **W1** and the other wire **W2**, the two wires **W** are fed in a state of being arranged in parallel with each other.

(95) By switching the forward and backward directions of the rotation direction of the feed motor **33a**, the wire feeding unit **3A** switches the direction of rotation of the first feed gear **30L** and the direction of rotation of the second feed gear **30R**, and the forward and reverse of the feeding direction of the wire **W** are switched.

(96) In the reinforcing bar binding machine **1A**, by forward rotation of the first feed gear **30L** and the second feed gear **30R** in the wire feeding unit **3A**, the wire **W** is fed in the forward direction indicated by the arrow **X1**, that is, in the direction of the curl guide unit **5A** and is wound around the reinforcing bar **S** at the curl guide unit **5A**. Further, after the wire **W** is wound around the reinforcing bar **S**, the first feed gear **30L** and the second feed gear **30R** are reversely rotated, whereby the wire **W** is fed in the backward direction indicated by the arrow **X2**, that is, in the direction of the magazine **2A** (pulled back). The wire **W** is wound around the reinforcing bar **S** and then pulled back, whereby the wire **W** is brought into close contact with the reinforcing bar **S**.

(97) FIGS. **4A**, **4B**, **4C**, and **4D** are views illustrating an example of the displacement unit of the present embodiment. The displacement unit **34** is an example of a displacement unit, and includes a first displacement member **35** that displaces the second feed gear **30R** in a direction in which the second feed gear **30R** is brought into close contact and separated with/from the first feed gear **30L** and a second displacement member **36** that displaces the first displacement member **35** in the rotation operation with the shaft **34a** illustrated in FIG. **2** as a fulcrum. The second feed gear **30R** is pressed in the direction of the first feed gear **30L** by a spring **37** that biases the second displacement member **36** that is displaced by a rotational operation with the shaft **36a** as a fulcrum. Thus, in this example, the two wires **W** are held between the first feed groove **32L** of the first feed gear **30L** and the second feed groove **32R** of the second feed gear **30R**. Further, the tooth portion **31L** of the first feed gear **30L** and the tooth portion **31R** of the second feed gear **30R** mesh with each other. Here, the relationship between the first displacement member **35** and the second displacement member **36** is such that by displacing the second displacement member **36** to bring the first displacement member **35** into a free state, the second feed gear **30R** can be separated from the first feed gear **30L**. However, the first displacement member **35** and the second displacement member **36** may be interlocked with each other.

(98) The displacement unit **34** includes an operation button **38** for pressing the second displacement member **36** and a release lever **39** for locking and unlocking the operation button **38**. The operation button **38** is an example of an operation member, protrudes outward from the main body **10A**, and is supported so as to be movable in directions indicated by arrows **T1** and **T2**.

(99) The operation button **38** has, as a locking mechanism, a first locking recess **38a** to which the release lever **39** is locked at a wire feed position where the wire **W** can be fed by the first feed gear **30L** and the second feed gear **30R**, and a second locking recess **38b** to which the release lever **39** is locked at a wire loading position where the wire **W** can be loaded by separating the first feed gear **30L** and the second feed gear **30R**.

(100) The release lever **39** is an example of a release member and is supported so as to be movable in directions indicated by arrows **U1** and **U2** intersecting the movement direction of the operation button **38**. The release lever **39** includes, as a locking mechanism, a locking protrusion **39a** to be locked to the first locking recess **38a** and the second locking recess **38b** of the operation button **38**.

(101) The release lever **39** is biased by a spring **39b** in the direction of the arrow **U1** approaching

the operation button **38** and is locked such that the locking protrusion **39a** enters the first locking recess **38a** of the operation button **38** in the wire feed position shown in FIG. 5A, or the locking protrusion **39a** enters the second locking recess **38b** of the operation button **38** in the wire loading position shown in FIG. 5B.

(102) A guide slope **39c** along the movement direction of the operation button **38** is formed on the locking protrusion **39a**. In the release lever **39**, the guide slope **39c** is pushed by the operation in which the operation button **38** at the wire feed position is pushed in the direction of the arrow T2, and the locking protrusion **39a** disengages from the first locking recess **38a**, whereby the release lever **39** is displaced moves in a direction of the arrow U2.

(103) The displacement unit **34** includes the second displacement member **36** in a direction substantially orthogonal to the feeding direction of the wire W fed by the first feed gear **30L** and the second feed gear **30R** in the wire feeding unit **3A**, behind the first feed gear **30L** and the second feed gear **30R**, that is, on the side of the handle unit **11A** with respect to the wire feeding unit **3A** in the main body **10A**. Also, the operation button **38** and the release lever **39** are provided behind the first feed gear **30L** and the second feed gear **30R**, that is, on the handle unit **11A** side with respect to the wire feeding unit **3A** in the main body **10A**.

(104) As illustrated in FIG. 4A, when the operation button **38** is in the wire feed position, the locking protrusion **39a** of the release lever **39** is locked to the first locking recess **38a** of the operation button **38**, and the operation button **38** is held at the wire feed position.

(105) As illustrated in FIG. 4A, in the displacement unit **34**, when the operation button **38** is in the wire feed position, the second displacement member **36** is pressed by the spring **37**, and the second displacement member **36** rotates about the shaft **36a** as a fulcrum, and is displaced in a direction pressing against the first feed gear **30L**.

(106) As illustrated in FIG. 4B, in the displacement unit **34**, when the operation button **38** is in the wire loading position, the locking protrusion **39a** of the release lever **39** is locked to the second locking recess **38b** of the operation button **38** and the operation button **38** is held at the wire loading position.

(107) As illustrated in FIG. 4B, in the displacement unit **34**, when the operation button **38** is in the wire loading position, the second displacement member **36** is pressed by the operation button **38** and the second displacement member **36** displaces the second feed gear **30R** in a direction away from the first feed gear **30L** with the shaft **36a** as a fulcrum.

(108) FIGS. 5A, 5B, and 5C are views illustrating an example of a parallel guide according to the present embodiment. Here, FIGS. 6A, 6B, and 6C are cross-sectional views taken along the line C-C of FIG. 2 and show the cross sectional shape of the parallel guide **4A** provided at the introduction position P1. Further, the cross-sectional view taken along a line D-D of FIG. 2 illustrating the sectional shape of the parallel guide **4A** provided at the intermediate position P2, and the cross-sectional view taken along a line E-E of FIG. 2 illustrating the sectional shape of the parallel guide **4A** provided at the cutting discharge position P3 show the same shape. Further, FIG. 5D is a view illustrating an example of parallel wires, and FIG. 5E is a view illustrating an example of twisted wires intersecting each other.

(109) The parallel guide **4A** is an example of a restricting unit constituting the feeding unit and restricts the direction of a plurality of (two or more) wires W that have been sent. Two or more wires W enter and the parallel guide **4A** feeds the two or more wires W in parallel. In the parallel guide **4A**, two or more wires are arranged in parallel along a direction orthogonal to the feeding direction of the wire W. Specifically, two or more wires W are arranged in parallel along the axial direction of the loop-like wire W wound around the reinforcing bar S by the curl guide unit **5A**. The parallel guide **4A** has a wire restricting unit (for example, an opening **4AW** described later) that restricts the directions and relative movement of the two or more wires W and makes them parallel. In this example, the parallel guide **4A** has a guide main body **4AG**, and the guide main body **4AG** is formed with an opening **4AW** which is the wire restricting unit for passing (inserting) a plurality

of wires W. The opening 4AW penetrates the guide main body 4AG along the feeding direction of the wire W. When the plurality of sent wires W pass through the opening 4AW and after passing through the opening 4AW, the shape thereof is determined so that the plurality of wires W are arranged in parallel (that is, each of the plurality of wires W is aligned in a direction (radial direction) orthogonal to the feeding direction of the wire W (axial direction) and the axis of each of the plurality of wires W is substantially parallel to each other). Therefore, the plurality of wires W that have passed through the parallel guide 4A go out from the parallel guide 4A in a state of being arranged in parallel. In this way, the parallel guide 4A restricts the direction and orientation in which the two wires W are aligned in the radial direction so that the two wires W are arranged in parallel. Therefore, in the opening 4AW, one direction orthogonal to the feeding direction of the wire W is longer than the other direction which is orthogonal to the feeding direction of the wire W orthogonal to the one direction. The opening 4AW has a longitudinal direction (in which two or more wires W can be juxtaposed) is disposed along a direction orthogonal to the feeding direction of the wire W, more specifically, along the axial direction of the wire W loop-shaped by the curl guide unit 5A. As a result, two or more wires W inserted through the opening 4AW are fed in parallel to the feeding direction of the wire W, and an axis of one wire is offset from an axis of the other wire in a direction parallel to the axial direction R_{ut} of the loop of wire W.

(110) In the following description, when describing the shape of the opening 4AW, a cross-sectional shape (along a cross-section cut in a direction orthogonal to the feeding direction, and viewed in the feeding direction of the wire W) will be described. The cross-sectional shape in the direction along the feeding direction of the wire W will be described in each case.

(111) For example, when the opening 4AW (the cross section thereof) is a circle having a diameter equal to or more than twice of the diameter of the wire W, or the length of one side is substantially a square which is twice or more the diameter of the wire W, the two wires W passing through the opening 4AW are in a state where they can freely move in the radial direction.

(112) If the two wires W passing through the opening 4AW can freely move in the radial direction within the opening 4AW, the direction in which the two wires W are arranged in the radial direction cannot be restricted, whereby the two wires W coming out from the opening 4AW may not be in parallel, may be twisted or intersected.

(113) In view of this, the opening 4AW is formed such that the length in the one direction, that is, the length $L1$ in the longitudinal direction is set to be slightly (n) times longer than the diameter r of the wire W in the form in which the plurality (n) of wires W are arranged along the radial direction, and the length in the other direction, that is, the length $L2$ in the lateral direction is set to be slightly (n) times longer than the diameter r of one wire W. In the present example, the opening 4AW has a length $L1$ in the longitudinal direction slightly twice longer than a diameter r of the wire W, and a length $L2$ in the lateral direction slightly longer than a diameter r of one wire W. In the present embodiment, the parallel guide 4A is configured such that the longitudinal direction of the opening 4AW is linear and the lateral direction is arcuate, but the configuration is not limited thereto.

(114) In the example illustrated in FIG. 5A, the length $L2$ in the lateral direction of the parallel guide 4A is set to a length slightly longer than the diameter r of one wire W as a preferable length. However, since it is sufficient that the wire W comes off from the opening 4AW in a parallel state without intersecting or being twisted, in the configuration in which the longitudinal direction of the parallel guide 4A is oriented along the axial direction R_{u1} of the loop of the wire W wound around the reinforcing bar S at the curl guide unit 5A, the length $L2$ of the parallel guide 4A in the lateral direction, as illustrated in FIG. 5B, may be within a range from a length slightly longer than the diameter r of one wire W to a length slightly shorter than the diameter r of two wires W.

(115) Further, in the configuration in which the longitudinal direction of the parallel guide 4A is oriented in a direction orthogonal to the axial direction R_{u1} of the loop of the wire W around the reinforcing bar S in the curl guide unit 5A, as illustrated in FIG. 5C, the length $L2$ in the lateral

direction of the parallel guide **4A** may be within a range from a length slightly longer than the diameter r of one wire W to a length shorter than the diameter r of two wires W .

(116) In the parallel guide **4A**, the longitudinal direction of the opening **4AW** is oriented along a direction orthogonal to the feeding direction of the wire W , in this example, along the axial direction $Ru1$ of the loop of the wire W wound around the reinforcing bar S in the curl guide unit **5A**.

(117) As a result, the parallel guide **4A** can pass two wires in parallel along the axial direction $Ru1$ of the loop of the wire W .

(118) In the parallel guide **4A**, when the length $L2$ in the lateral direction of the opening **4AW** is shorter than twice the diameter r of the wire W and slightly longer than the diameter r of the wire W , even if the length $L1$ in the longitudinal direction of the opening **4AW** is sufficiently twice or more times longer than the diameter r of the wire W , it is possible to pass the wires W in parallel.

(119) However, the longer the length $L2$ in the lateral direction (for example, the length close to twice the diameter r of the wire W) and the longer the length $L1$ in the longitudinal direction, the wire W can further freely move in the opening **4AW**. Then, the respective axes of the two wires W do not become parallel in the opening **4AW**, and there is a high possibility that the wires W are twisted or intersect each other after passing through the opening **4AW**.

(120) Therefore, it is preferable that the longitudinal length $L1$ of the opening **4AW** is slightly longer than twice the diameter r of the wire W , and the length $L2$ in the lateral direction is also slightly longer than the diameter r of the wire W so that the two wires W are arranged in parallel in the feed direction, and are adjacent each other in the lateral or radial direction.

(121) The parallel guide **4A** is provided at predetermined positions on the upstream side and the downstream side of the first feed gear **30L** and the second feed gear **30R** (the wire feeding unit **3A**) with respect to the feeding direction for feeding the wire W in the forward direction. By providing the parallel guide **4A** on the upstream side of the first feed gear **30L** and the second feed gear **30R**, the two wires W in a parallel state enter the wire feeding unit **3A**. Therefore, the wire feeding unit **3A** can feed the wire W appropriately (in parallel). Furthermore, by providing the parallel guide **4A** also on the downstream side of the first feed gear **30L** and the second feed gear **30R**, while maintaining the parallel state of the two wires W sent from the wire feeding unit **3A**, the wire W can be further sent to the downstream side.

(122) The parallel guides **4A** provided on the upstream side of the first feed gear **30L** and the second feed gear **30R** are provided at the introduction position **P1** between the first feed gear **30L** and the second feed gear **30R** and the magazine **2A** such that the wires W fed to the wire feeding unit **3A** are arranged in parallel in a predetermined direction.

(123) One of the parallel guides **4A** provided on the downstream side of the first feed gear **30L** and the second feed gear **30R** is provided at the intermediate position **P2** between the first feed gear **30L** and the second feed gear **30R** and the cutting unit **6A** such that the wires W fed to the cutting unit **6A** are arranged in parallel in the predetermined direction.

(124) Further, the other one of the parallel guides **4A** provided on the downstream side of the first feed gear **30L** and the second feed gear **30R** is provided at the cutting discharge position **P3** where the cutting unit **6A** is disposed such that the wires W fed to the curl guide unit **5A** are arranged in parallel in the predetermined direction.

(125) The parallel guide **4A** provided at the introduction position **P1** has the above-described shape in which at least the downstream side of the opening **4AW** restricts the radial direction of the wire W with respect to the feeding direction of the wire W sent in the forward direction. On the other hand, the opening area of the side facing the magazine **2A** (the wire introducing unit), which is the upstream side of the opening **4AW** with respect to the feeding direction of the wire W sent in the forward direction, has a larger opening area than the downstream side. Specifically, the opening **4AW** has a tube-shaped hole portion that restricts the direction of the wire W and a conical (funnel-shaped, tapered) hole portion in which an opening area gradually increases from the upstream side

end of the tube-shaped hole portion to the inlet portion of the opening 4AW as the wire introducing portion. By making the opening area of the wire introducing portion the largest and gradually reducing the opening area therefrom, it is easy to allow the wire W to enter the parallel guide 4. Therefore, the work of introducing the wire W into the opening 4AW can be performed easily. (126) The other parallel guide 4A also has the same configuration, and the downstream opening 4AW with respect to the feeding direction of the wire W sent in the forward direction has the above-described shape that restricts the direction of the wire W in the radial direction. Further, with regard to the other parallel guide 4, the opening area of the opening on the upstream side with respect to the feeding direction of the wire W sent in the forward direction may be made larger than the opening area of the opening on the downstream side.

(127) The parallel guide 4A provided at the introduction position P1, the parallel guide 4A provided at the intermediate position P2, and the parallel guide 4A provided at the cutting discharge position P3 are arranged such that the longitudinal direction of the opening 4AW orthogonal to the feeding direction of the wire W is in the direction along the axial direction Ru1 of the loop of wire W wound around the reinforcing bar S.

(128) As a result, as illustrated in FIG. 5D, the two wires W sent by the first feed gear 30L and the second feed gear 30R are sent while maintaining a state of being arranged in parallel in the axial direction Ru1 of the loop of wire W wound around the reinforcing bar S, and, as illustrated in FIG. 5E, the two wires W are prevented from intersecting and being twisted during feeding.

(129) In the present example, the opening 4AW is a tube-shaped hole having a predetermined depth (a predetermined distance or depth from the inlet to the outlet of the opening 4AW) from the inlet to the outlet of the opening 4AW (in the feeding direction of the wire W), but the shape of the opening 4AW is not limited to this. For example, the opening 4AW may be a planar hole having almost no depth with which the plate-like guide main body 4AG is opened. Further, the opening 4AW may be a groove-shaped guide (for example, a U-shaped guide groove with an opened upper portion) instead of the hole portion penetrating through the guide main body 4AG. Furthermore, in the present example, the opening area of the inlet portion of the opening 4AW as the wire introducing portion is made larger than the other portion, but it may not necessarily be larger than the other portion. The shape of the opening 4AW is not limited to a specific shape as long as the plurality of wires that have passed through the opening 4AW and come out of the parallel guide 4A are in a parallel state.

(130) Hitherto, an example in which the parallel guide 4A is provided at the upstream side (introduction position P1) and a predetermined position (intermediate position P2 and cutting discharge position P3) on the downstream side of the first feed gear 30L and the second feed gear 30R is described. However, the position where the parallel guide 4A is installed is not necessarily limited to these three positions. That is, the parallel guide 4A may be installed only in the introduction position P1, only in the intermediate position P2, or only in the cutting discharge position P3, and only in the introduction position P1 and the intermediate position P2, only in the introduction position P1 and the cutting discharge position P3, or only in the intermediate position P2 and the cutting discharge position P3. Further, four or more parallel guides 4A may be provided at any position between the introduction position P1 and the curl guide unit 5A on the downstream side of the cutting position P3. The introduction position P1 also includes the inside of the magazine 2A. That is, the parallel guide 4A may be arranged in the vicinity of the outlet from which the wire W is drawn inside the magazine 2A.

(131) The curl guide unit 5A is an example of guide unit constituting the feeding unit and constitutes a conveying path for winding the two wires W around the reinforcing bars S in a loop shape. The curl guide unit 5A includes a first guide unit 50 for curling the wire W sent by the first feed gear 30L and the second feed gear 30R and a second guide unit 51 for guiding the wire W fed from the first guide unit 50 to the binding unit 7A.

(132) The first guide unit 50 includes guide grooves 52 constituting a feed path of the wire W and

guide pins **53** and **53b** as a guide member for curling the wire **W** in cooperation with the guide groove **52**. FIG. **6** is a view illustrating an example of the guide groove of the present embodiment. FIG. **6** is a sectional view taken along the line G-G of FIG. **2**.

(133) The guide groove **52** forms a guide unit and restricts a direction in the radial direction of movement the wire **W** orthogonal to the feeding direction of the wire **W** together with the parallel guide **4A**. Therefore, in this example, the guide groove **52** is configured by an opening with an elongated shape in which one direction orthogonal to the feeding direction of the wire **W** is longer than the other direction orthogonal to the feeding direction of the wire **W** and orthogonal to the one direction.

(134) The guide groove **52** has a longitudinal length **L1**, which is a length of the width direction of the groove, slightly longer than the diameter **r** of a plurality of wires **W** in a form in which the wires **W** are arranged along the radial direction and a lateral length **L2** slightly longer than the diameter **r** of one wire **W**. In the present embodiment, the length **L1** of the guide groove **52** in the longitudinal direction is slightly longer than the diameter **r** of two wires **W**. In the guide groove **52**, the longitudinal direction of the opening is arranged in the direction along the axial direction **Ru1** of the loop of wire **W**. It should be noted that the guide groove **52** may not necessarily have the function of restricting the direction of the wire **W** in the radial direction. In that case, the length (length) in the longitudinal direction and in the lateral direction of the guide groove **52** is not limited to the above-described size.

(135) The guide pin **53** is provided on the side of the introducing portion of the wire **W** that is fed by the first feed gear **30L** and the second feed gear **30R** in the first guide unit **50** and is arranged inside the loop **Ru** formed by the wire **W** in the radial direction with respect to the feed path of the wire **W** by the guide groove **52**. The guide pin **53** restricts the feed path of the wire **W** so that the wire **W** fed along the guide groove **52** does not enter the inside of the loop **Ru** formed by the wire **W** in the radial direction.

(136) The guide pin **53b** is provided on the side of the discharge portion of the wire **W** which is fed by the first feed gear **30L** and the second feed gear **30R** in the first guide unit **50** and is arranged on the outer side in the radial direction of the loop **Ru** formed by the wire **W** with respect to the feed path of the wire **W** by the guide groove **52**.

(137) In the wire **W** sent by the first feed gear **30L** and the second feed gear **30R**, the radial position of the loop **Ru** formed by the wire **W** is restricted at least at three points including two points on the outer side in the radial direction of the loop **Ru** formed by the wire **W** and at least one point on the inner side between the two points, so that the wire **W** is curled.

(138) In this example, the radially outer position of the loop **Ru** formed by the wire **W** is restricted at two points of the parallel guide **4A** at the cutting discharge position **P3** provided on the upstream side of the guide pin **53** with respect to the feeding direction of the wire **W** sent in the forward direction and the guide pin **53b** provided on the downstream side of the guide pin **53**. Further, the radially inner position of the loop **Ru** formed by the wire **W** is restricted by the guide pin **53**.

(139) The curl guide unit **5A** includes a retreat mechanism **53a** for allowing the guide pin **53** to retreat from a path through which the wire **W** moves by an operation of winding the wire **W** around the reinforcing bar **S**. After the wire **W** is wound around the reinforcing bar **S**, the retreat mechanism **53a** is displaced in conjunction with the operation of the binding unit **7A**, and retreats the guide pin **53** from the path where the wire **W** moves before the timing of winding the wire **W** around the reinforcing bar **S**.

(140) The second guide unit **51** includes a fixed guide unit **54** as a third guide unit for restricting the radial position of the loop **Ru** (movement of the wire **W** in the radial direction of the loop **Ru**) formed by the wire **W** wound around the reinforcing bar **S** and a movable guide unit **55** serving as a fourth guide unit for restricting the position along the axial direction **Ru1** of the loop **Ru** formed by the wire **W** wound around the reinforcing bar **S** (movement of the wire **W** in the axial direction **Ru1** of the loop **Ru**).

(141) FIGS. 7, 8A, 8B, 9A, and 9B are views illustrating an example of a second guide unit, FIG. 7 is a plan view of the second guide unit 51 as viewed from above, FIGS. 8A and 8B are side views of the second guide unit 51 as viewed from one side, and FIGS. 9A and 9B are side views of the second guide unit 51 as viewed from the other side.

(142) The fixed guide unit 54 is provided with a wall surface 54a as a surface extending along the feeding direction of the wire W on the outer side in the radial direction of the loop Ru formed by the wire W wound around the reinforcing bar S. When the wire W is wound around the reinforcing bar S, the wall surface 54a of the fixed guide unit 54 restricts the radial position of the loop Ru formed by the wire W wound around the reinforcing bar S. The fixed guide unit 54 is fixed to the main body 10A of the reinforcing bar binding machine 1A, and the position thereof is fixed with respect to the first guide unit 50. The fixed guide unit 54 may be integrally formed with the main body 10A. In addition, in the configuration in which the fixed guide unit 54, which is a separate component, is attached to the main body 10A, the fixed guide unit 54 is not perfectly fixed to the main body 10A, but in the operation of forming the loop Ru may be movable to such an extent that movement of the wire W can be restricted.

(143) The movable guide unit 55 includes a wall surface 55a that is provided on the distal end side of the second guide unit 51, and the wall surface 55a is provided on both sides along the axial direction Ru1 of the loop Ru formed by the wire W wound around the reinforcing bar S and is erected inward in the radial direction of the loop Ru from the wall surface 54a toward the inside is provided. When the wire W is wound around the reinforcing bar S, the movable guide unit 55 restricts the position along the axial direction Ru1 of the loop Ru formed by the wire W wound around the reinforcing bar S using the wall surface 55a. The wall surface 55a of the movable guide unit 55 has a tapered shape in which the gap of the wall surfaces 55a is spread at the distal end side where the wire W sent from the first guide unit 50 enters and narrows toward the fixed guide unit 54b. As a result, the position of the wire W sent from the first guide unit 50 in the axial direction Ru1 of the loop Ru wound around the reinforcing bar S is restricted by the wall surface 55a of the movable guide unit 55, and guided to the fixed guide unit 54 by the movable guide unit 55.

(144) The movable guide unit 55 is supported on the fixed guide unit 54 by a shaft 55b on the side opposite to the tip side into which the wire W sent from the first guide unit 50 enters. In the movable guide unit 55, the distal end side thereof into which the wire W fed from the first guide unit 50 enters is opened and closed in the direction to come into contact with and separate from the first guide unit 50 by the rotation operation of the loop Ru formed by the wire W wound around the reinforcing bar S along the axial direction Ru1 with the shaft 55b as a fulcrum.

(145) In the reinforcing bar binding machine, when binding the reinforcing bar S, between a pair of guide members provided for winding the wire W around the reinforcing bar S, in this example, between the first guide unit 50 and the second guide unit 51, a reinforcing bar is inserted (set) and then the binding work is performed. When the binding work is completed, in order to perform the next binding work, the first guide unit 50 and the second guide unit 51 are pulled out from the reinforcing bar S after the completion of the binding. In the case of pulling out the first guide unit 50 and the second guide unit 51 from about the reinforcing bar S, if the reinforcing bar binding machine 1A is moved in the direction of the arrow Z3 (see FIG. 1) which is one direction of separation from the reinforcing bar S, the reinforcing bar S can be pulled out from the first guide unit 50 and the second guide unit 51 without any problem. However, for example, when the reinforcing bar S is arranged at a predetermined interval along the arrow Y2 and these reinforcing bars S are sequentially bound, moving the reinforcing bar binding machine 1A in the direction of the arrow Z3 after each binding is troublesome, and if it can be moved in the direction of arrow Z2, the binding work can be performed quickly. However, in the conventional reinforcing bar binding machine disclosed in, for example, Japanese Patent No. 4747456, since the guide member corresponding to the second guide unit 51 in the present example is fixed to the binding machine body, when trying to move the reinforcing bar binding machine in the direction of the arrow Z2, the

guide member is caught on the reinforcing bar S. Therefore, in the reinforcing bar binding machine **1A**, the second guide unit **51** (the movable guide unit **55**) is made movable as described above and the reinforcing bar binding machine **1A** is moved in the direction of the arrow **Z2** so that the reinforcing bar S is more easily pulled out from between the first guide unit **50** and the second guide unit **51**.

(146) Therefore, the movable guide unit **55** rotates about the shaft **55b** as a fulcrum, and thus opened and closed between a guide position at which the wire W sent out from the first guide unit **50** can be guided to the second guide unit **51** and a retreat position at which the reinforcing bar binding machine **1A** is moved in the direction of the arrow **Z2** and then is retreated by the operation of pulling out the reinforcing bar binding machine **1A** from the reinforcing bar S.

(147) The movable guide unit **55** is urged in a direction in which the distance between the distal end side of the first guide unit **50** and the distal end side of the second guide unit **51** is reduced by the urging unit (biasing unit) such as a torsion coil spring **57**, and is held in the guide position illustrated in FIGS. **8A** and **9A** by the force of the torsion coil spring **57**. In addition, when the movable guide unit **55** is pushed to the reinforcing bar S by the operation of pulling out the reinforcing bar binding machine **1A** from the reinforcing bar S, the movable guide unit **55** is opened from the guide position to the retreat position illustrated in FIGS. **8B** and **9B**. The guide position is a position where the wall surface **55a** of the movable guide unit **55** exists at a position where the wire W forming the loop Ru passes. The retreat position is a position at which the reinforcing bar S presses the movable guide unit **55** by the movement of the reinforcing bar binding machine **1A**, and the reinforcing bar S can be pulled out from between the first guide unit **50** and the second guide unit **51**. Here, the direction in which the reinforcing bar binding machine **1A** is moved is not uniform, and even if the movable guide unit **55** slightly moves from the guide position, the reinforcing bar S can be pulled out from between the first guide unit **50** and the second guide unit **51**, and thus a position slightly moved from the guide position is also included in the retreat position.

(148) The reinforcing bar binding machine **1A** includes a guide opening/closing sensor **56** that detects opening and closing of the movable guide unit **55**. The guide opening/closing sensor **56** detects the closed state and the open state of the movable guide unit **55**, and outputs a predetermined detection signal.

(149) The cutting unit **6A** includes a fixed blade unit **60**, a rotary blade unit **61** for cutting the wire W in cooperation with the fixed blade unit **60**, and a transmission mechanism **62** which transmits the operation of the binding unit **7A**, in this example, the operation of a movable member **83** (to be described later) moving in a linear direction to the rotary blade unit **61** and rotates the rotary blade unit **61**. The fixed blade unit **60** is configured by providing an edge portion capable of cutting the wire W in the opening through which the wire W passes. In the present example, the fixed blade unit **60** includes a parallel guide **4A** arranged at the cutting discharge position **P3**.

(150) The rotary blade unit **61** cuts the wire W passing through the parallel guide **4A** of the fixed blade unit **60** by the rotation operation with the shaft **61a** as a fulcrum. The transmission mechanism **62** is displaced in conjunction with the operation of the binding unit **7A**, and after the wire W is wound around the reinforcing bar S, the rotary blade unit **61** is rotated according to the timing of twisting the wire W to cut the wire W.

(151) The binding unit **7A** is an example of a binding unit, and includes a gripping unit **70** that grips the wire W and a bending unit **71** configured to bend one end WS side and the other end WE side of the wire W gripped by the gripping unit **70** toward the reinforcing bar S.

(152) The gripping unit **70** is an example of a gripping unit, and includes a fixed gripping member **70C**, a first movable gripping member **70L**, and a second movable gripping member **70R** as illustrated in FIG. **2**. The first movable gripping member **70L** and the second movable gripping member **70R** are arranged in the lateral direction via the fixed gripping member **70C**. Specifically, the first movable gripping member **70L** is disposed on one side along the axial direction of the wire

W to be wound around, with respect to the fixed gripping member **70C**, and the second movable gripping member **70R** is disposed on the other side.

(153) The first movable gripping member **70L** is displaced in a direction to come into contact with and separate from the fixed gripping member **70C**. In addition, the second movable gripping member **70R** is displaced in a direction to come into contact with and separate from the fixed gripping member **70C**.

(154) As the first movable gripping member **70L** moves in a direction away from the fixed gripping member **70C**, in the gripping unit **70**, a feed path through which the wire W passes between the first movable gripping member **70L** and the fixed gripping member **70C** is formed. On the other hand, as the first movable gripping member **70L** moves toward the fixed gripping member **70C**, the wire W is gripped between the first movable gripping member **70L** and the fixed gripping member **70C**.

(155) When the second movable gripping member **70R** moves in a direction away from the fixed gripping member **70C**, in the gripping unit **70** forms a feed path through which the wire W passes between the second movable gripping member **70R** and the fixed gripping member **70C** is formed. On the other hand, as the second movable gripping member **70R** moves toward the fixed gripping member **70C**, the wire W is gripped between the second movable gripping member **70R** and the fixed gripping member **70C**.

(156) The wire W sent by the first feed gear **30L** and the second feed gear **30R** and passed through the parallel guide **4A** at the cutting discharge position **P3** passes between the fixed gripping member **70C** and the second movable gripping member **70R** and is guided to the curl guide unit **5A**. The wire W which has been wound by the curl guide unit **5A** passes between the fixed gripping member **70C** and the first movable gripping member **70L**.

(157) Therefore, a first gripping unit for gripping one end WS side of the wire W is constituted by a pair of gripping members, for example, the fixed gripping member **70C** and the first movable gripping member **70L**. Further, the fixed gripping member **70C** and the second movable gripping member **70R** constitute a second gripping unit for gripping the other end WE side of the wire W cut by the cutting unit **6A**.

(158) FIGS. **10A** and **19B** are views illustrating main parts of the gripping unit of this embodiment. The first movable gripping member **70L** includes a protrusion **70Lb** protruding toward the fixed gripping member **70C** on a surface facing the fixed gripping member **70C**. On the other hand, the fixed gripping member **70C** includes a recess **73**, into which the protrusion **70Lb** of the first movable gripping member **70L** is inserted, on a surface facing the first movable gripping member **70L**. Accordingly, when the wire W is gripped with the first movable gripping member **70L** and the fixed gripping member **70C**, the wire W is bent toward the first movable gripping member **70L**.

(159) Specifically, the fixed gripping member **70C** includes a preliminary bending portion **72**. The preliminary folding portion **72** is configured such that a protrusion protruding toward the first movable gripping member **70L** is provided at a downstream end along the feeding direction of the wire W fed in the forward direction on the surface facing the first movable gripping member **70L** of the fixed gripping member **70C**.

(160) In order to grip the wire W between the fixed gripping member **70C** and the first movable gripping member **70L** and prevent the gripped wire W from being pulled out, the gripping unit **70** has the protrusion portion **72b** and the recess portion **73** on the fixed gripping member **70C**. The protrusion portion **72b** is provided on the upstream end along the feeding direction of the wire W fed in the forward direction on the surface facing the first movable gripping member **70L** of the fixed gripping member **70C** and protrudes to the first movable gripping member **70L**. The recess portion **73** is provided between the preliminary bending portion **72** and the protrusion portion **72b** and has a recess shape in a direction opposite to the first movable gripping member **70L**.

(161) The first movable gripping member **70L** has a recess portion **70La** into which the preliminary bending portion **72** of the fixed gripping member **70C** enters and a protrusion portion **70Lb** which

enters the recess portion 73 of the fixed gripping member 70C.

(162) As a result, as illustrated in FIG. 10B, by the operation of gripping one end WS side of the wire W between the fixed gripping member 70C and the first movable gripping member 70L, the wire W is pressed by the preliminary bending portion 72 on the first movable gripping member 70L side, and one end WS of the wire W is bent in a direction away from the wire W gripped by the fixed gripping member 70C and the second movable gripping member 70R.

(163) Gripping the wire W with the fixed gripping member 70C and the second movable gripping member 70R includes a state in which the wire W can move freely to some extent between the fixed gripping member 70C and the second movable gripping member 70R. This is because, in the operation of winding the wire W around the reinforcing bar S, it is necessary to move the wire W between the fixed gripping member 70C and the second movable gripping member 70R.

(164) The bending portion 71 is an example of a bending unit which bends the wire W such that the end of the wire W is located closer to the binding object than a top of the wire W protruding most in a direction away from the binding object after binding of the binding object. The bending portion 71 bends the wire W gripped by the gripping unit 70 before the wire W is twisted by the gripping unit 70.

(165) The bending portion 71 is provided around the gripping unit 70 so as to cover a part of the gripping unit 70, and is provided so as to be movable along the axial direction of the gripping unit 70. Specifically, the bending portion 71 approaches one end WS side of the wire W gripped by the fixed gripping member 70C and the first movable gripping member 70L and the other end WE side of the wire W gripped by the fixed gripping member 70C and the second movable gripping member 70R, and is horizontally movable in the direction in which one end WS side and the other end WE side of the wire W are bent and away from the bent wire W.

(166) The bending portion 71 moves in the forward direction (see FIG. 1) indicated by an arrow F, so that one end WS side of the wire W gripped by the fixed gripping member 70C and the first movable gripping member 70L is bent to the reinforcing bar S side with the gripping position as the fulcrum. Further, the bending portion 71 moves in the forward direction indicated by the arrow F, whereby the other end WE side of the wire W between the fixed gripping member 70C and the second movable gripping member 70R is bent to the reinforcing bar S side with the gripping position as the fulcrum.

(167) The wire W is bent by the movement of the bending portion 71, so that the wire W passing between the second movable gripping member 70R and the fixed gripping member 70C is pressed by the bending portion 71, and the wire W is prevented from coming off between the fixed gripping member 70C and the second movable gripping member 70R.

(168) The binding unit 7A includes a length restricting unit 74 that restricts the position of one end WS of the wire W. The length restricting unit 74 is constituted by providing a member against which the one end WS of the wire W abuts in the feed path of the wire W that has passed between the fixed gripping member 70C and the first movable gripping member 70L. In order to secure a predetermined distance from the gripping position of the wire W by the fixed gripping member 70C and the first movable gripping member 70L, the length restricting unit 74 is provided in the first guide unit 50 of the curl guide unit 5A in this example.

(169) The reinforcing bar binding machine 1A includes a binding unit driving mechanism 8A that drives the binding unit 7A. The binding unit driving mechanism 8A includes a motor 80, a rotary shaft 82 driven by the motor 80 via a speed reducer 81 that performs deceleration and torque amplification, a movable member 83 that is displaced by a rotation operation of the rotary shaft 82, and a rotation restricting member 84 that restricts the rotation of the movable member 83 interlocking with the rotation operation of the rotary shaft 82.

(170) In the rotary shaft 82 and the movable member 83, by the screw portion provided on the rotary shaft 82 and the nut portion provided in the movable member 83, the rotation operation of the rotary shaft 82 is converted to the movement of the movable member 83 along the rotary shaft

82 in the forward and backward direction.

(171) The movable member **83** is locked to the rotation restricting member **84** in the operation region where the wire **W** is gripped by the gripping unit **70**, and then the wire **W** is bent by the bending portion **71**, so that the movable member **83** moves in the forward and backward direction in a state where the rotation operation is restricted by the rotation restricting member **84**. Further, the movable member **83** is rotated by the rotation operation of the rotary shaft **82** by coming off from the locking of the rotation restricting member **84**.

(172) In this example, the movable member **83** is connected to the first movable gripping member **70L** and the second movable gripping member **70R** via a cam (not illustrated). The binding unit driving mechanism **8A** is configured that the movement of the movable member **83** in the forward and backward direction is converted into the operation of displacing the first movable gripping member **70L** in the direction to come into contact with and separate from the fixed gripping member **70C**, and the operation of displacing the second movable gripping member **70R** in the direction to come into contact with and separate from the fixed gripping member **70C**.

(173) Further, in the binding unit driving mechanism **8A**, the rotation operation of the movable member **83** is converted into the rotation operation of the fixed gripping member **70C**, the first movable gripping member **70L** and the second movable gripping member **70R**.

(174) Furthermore, in the binding unit driving mechanism **8A**, the bending portion **71** is provided integrally with the movable member **83**, so that the bending portion **71** moves in the forward and backward direction by the movement of the movable member **83** in the forward and backward direction.

(175) The retreat mechanism **53a** of the guide pin **53** is configured by a link mechanism that converts the movement of the movable member **83** in the forward and backward direction into displacement of the guide pin **53**. The transmission mechanism **62** of the rotary blade portion **61** is configured by a link mechanism that converts the movement of the movable member **83** in the forward and backward direction into the rotation operation of the rotary blade portion **61**.

(176) FIG. **11** is an external view illustrating an example of the reinforcing bar binding machine of the present embodiment. The reinforcing bar binding machine **1A** according to the present embodiment has a form used by a worker in hand and includes a main body **10A** and a handle portion **11A**. As illustrated in FIG. **1** and the like, the reinforcing bar binding machine **1A** incorporates a binding unit **7A** and a binding unit driving mechanism **8A** in the main body **10A** and has a curl guide unit **5A** at one end side of the main body **10A** in the longitudinal direction (first direction **Y1**). Further, the handle portion **11A** is provided so as to protrude from the other end side in the longitudinal direction of the main body **10A** to one direction (second direction **Y2**) substantially orthogonal (intersecting) with the longitudinal direction. Further, the wire feeding unit **3A** is provided on the side along the second direction **Y2** with respect to the binding unit **7A**, the displacement unit **34** is provided on the other side along the first direction **Y1** with respect to the wire feeding unit **3A**, that is, on the handle portion **11A** with respect to the wire feeding unit **3A** in the main body **10A**, and the magazine **2A** is provided on the side along the second direction **Y2** with respect to the wire feeding unit **3A**.

(177) Therefore, the handle portion **11A** is provided on the other side along the first direction **Y1** with respect to the magazine **2A**. In the following description, in the first direction **Y1** along the direction in which the magazine **2A**, the wire feeding unit **3A**, the displacement unit **34**, and the handle portion **11A** are arranged, the side on which the magazine **2A** is provided is called a front side, and the side on which the handle portion **11A** is provided is called a back side. In the displacement unit **34**, a second displacement member **36** is provided in a direction substantially orthogonal to the feeding direction of the wire **W** fed by the first feed gear **30L** and the second feed gear **30R** in the wire feeding unit **3A**, behind the first feed gear **30L** and the second feed gear **30R** of the wire feeding unit **3A**, and between the first feed gear **30L** and the second feed gear **30R** and the handle portion **11A**. An operation button **38** for displacing the second displacement member **36**,

a release lever **39** for releasing locking and unlocking of the operation button **38** are provided between the first feed gear **30L** and the second feed gear **30R** and the handle portion **11A**.

(178) It is noted that a release function for releasing locking and unlocking may be mounted on the operation button **38** for displacing the second displacement member **36** (also serving as a release lever). That is, the displacement unit **34** includes the second displacement member **36** for displacing the first feed gear **30L** and the second feed gear **30R** of the wire feeding unit **3A** toward and away from each other, and the operation button **38** which displaces the second displacement member **36** and protrudes outwardly from the main body **10A**, and is positioned between the wire feeding unit **3A** and the handle portion **11A** in the main body **10A**.

(179) In this manner, by providing the mechanism for displacing the second feed gear **30R**, between the second feed gear **30R** and the handle portion **11A**, behind the second feed gear **30R**, as illustrated in FIG. 2, a mechanism for displacing the second feed gear **30R** is not provided in the feed path of the wire **W** below the first feed gear **30L** and the second feed gear **30R**. In other words, the interior of the magazine **2A**, which forms the feed path of the wire **W**, below the first feed gear **30L** and the second feed gear **30R** can be used as the wire loading space **22** which is the space for loading the wire **W** into the wire feeding unit **3A**. That is, the wire loading space **22** for the wire feeding unit **3A** can be formed inside the magazine **2A**.

(180) A trigger **12A** is provided on the front side of the handle portion **11A**, and the control unit **14A** controls the feed motor **33a** and the motor **80** according to the state of the switch **13A** pressed by the operation of the trigger **12A**. Further, a battery **15A** is detachably attached to a lower portion of the handle portion **11A**.

(181) <Example of Operation of Reinforcing Bar Binding Machine in Present Embodiment>

(182) FIGS. 12 to 19 are diagrams for explaining the operation of the reinforcing bar binding machine **1A** according to the present embodiment, and FIGS. 20A, 20B, and 20C are diagrams for explaining the operation of winding the wire around the reinforcing bar. FIGS. 21A and 21B are explanatory views of the operation of forming a loop with a wire by the curl guide unit, and FIGS. 22A, 22B, and 22C are explanatory views of the operation of bending the wire. Next, with reference to the drawings, the operation of binding the reinforcing bar **S** with the wire **W** by the reinforcing bar binding machine **1A** of this embodiment will be described.

(183) In order to load the wire **W** wound around the reel **20** housed in the magazine **2A**, first, the operation button **38** in the wire feed position illustrated in FIG. 4A is pushed in the arrow **T2** direction. When the operation button **38** is pushed in the direction of the arrow **T2**, the guide slope **39c** of the release lever **39** is pushed, and the locking protrusion **39a** comes off from the first locking recess **38a**. As a result, the release lever **39** is displaced in the arrow **U2** direction.

(184) When the operation button **38** is pushed to the wire loading position, as illustrated in FIG. 4B, the release lever **39** is pushed by the spring **39b** in the direction of the arrow **U1**, and the locking protrusion **39a** is inserted into the second locking recess **38b** of the operation button **38** and is locked. Therefore, the operation button **38** is held at the wire loading position.

(185) When the operation button **38** is in the wire loading position, the second displacement member **36** is pressed by the operation button **38**, and the second displacement member **36** displaces the second feed gear **30R** about the shaft **36a** as a fulcrum in a direction away from the first feed gear **30L**. Therefore, the second feed gear **30R** is separated from the first feed gear **30L**, and the wire **W** can be inserted between the first feed gear **30L** and the second feed gear **30R**.

(186) After loading the wire **W**, as illustrated in FIG. 4C, by pushing the release lever **39** in the direction of the arrow **U2**, the locking protrusion **39a** comes off from the second locking recess **38b** of the operation button **38**. As a result, the second displacement member **36** is pressed by the spring **37**, and the second displacement member **36** is displaced in the direction to press the second feed gear **30R** against the first feed gear **30L** about the shaft **36a** as a fulcrum. Therefore, the wire **W** is pinched between the first feed gear **30L** and the second feed gear **30R**.

(187) When the operation button **38** is pushed in the direction of the arrow **T1** by the second

displacement member **36** and is displaced to the wire feed position as illustrated in FIG. **4A**, the locking protrusion **39a** of the release lever **39** is locked to the first locking recess **38a** of the operation button **38**, and the operation button **38** is held at the wire feed position.

(188) FIG. **12** illustrates the origin state after the loading of the wire, that is, the initial state in which the wire **W** has not yet been sent by the wire feeding unit **3A**. In the origin state, the distal end of the wire **W** stands by at the cutting discharge position **P3**. As illustrated in FIG. **20A**, the wire **W** waiting at the cutting discharge position **P3** is arranged in parallel in a predetermined direction by passing through the parallel guide **4A** (fixed blade portion **60**) in which the two wires **W** are provided at the cutting discharge position **P3**, in this example.

(189) In the wire **W** between the cutting discharge position **P3** and the magazine **2A**, the parallel guide **4A** at the intermediate position **P2** and the parallel guide **4A** at the introduction position **P1** are arranged in parallel in a predetermined direction by the first feed gear **30L** and the second feed gear **30R**.

(190) FIG. **13** illustrates a state in which the wire **W** is wound around the reinforcing bar **S**. When the reinforcing bar **S** is inserted between the first guide unit **50** and the second guide unit **51** of the curl guide unit **5A** and the trigger **12A** is operated, the feed motor **33a** is driven in the normal rotation direction, and thus the first feed gear **30L** rotates in forward direction and the second feed gear **30R** rotates in the forward direction while following the first feed gear **30L**.

(191) Therefore, the two wires **W** are fed in the forward direction by the frictional force generated between the first feed gear **30L** and the one wire **W1**, the frictional force generated between the second feed gear **30R** and the other wire **W2**, and the frictional force generated between the one wire **W1** and the other wire **W2**.

(192) Two wires **W** entering between the first feed groove **32L** of the first feed gear **30L** and the second feed groove **32R** of the second feed gear **30R**, and two wires **W** discharged from the first feed gear **30L** and the second feed gear **30R** are fed in parallel with each other in a predetermined direction by providing the parallel guides **4A** on the upstream side and the downstream side of the wire feeding unit **3A** with respect to the feeding direction of the wire **W** fed in the forward direction.

(193) When the wire **W** is fed in the forward direction, the wire **W** passes between the fixed gripping member **70C** and the second movable gripping member **70R** and passes through the guide groove **52** of the first guide unit **50** of the curl guide unit **5A**. As a result, the wire **W** is curled so as to be wound around the reinforcing bar **S**. The two wires **W** introduced into the first guide unit **50** are held in a state of being arranged in parallel by the parallel guide **4A** at the cutting discharge position **P3**. Further, since the two wires **W** are fed in a state of being pressed against the outer wall surface of the guide groove **52**, the wires **W** passing through the guide groove **52** are also held in a state of being arranged in parallel in a predetermined direction.

(194) As illustrated in FIG. **21A**, the wire **W** fed from the first guide unit **50** is restricted to move along the axial direction **Ru1** of the loop **Ru** formed by the wire to be wound therearound by the movable guide unit **55** of the second guide unit **51**, to be guided to the fixed guide unit **54** by the wall surface **55a**. In FIG. **21B**, the movement of the wire **W** along the radial direction of the loop **Ru**, which is guided to the fixed guide unit **54**, is restricted by the wall surface **54a** of the fixed guide unit **54**, and the wire **W** is guided between the fixed gripping member **70C** and the first movable gripping member **70L**. Then, when the distal end of the wire **W** is fed to a position where it abuts against the length restricting unit **74**, driving of the feed motor **33a** is stopped.

(195) A slight amount of wire **W** is fed in the forward direction until the distal end of the wire **W** abuts against the length restricting unit **74** and then the feeding is stopped, whereby the wire **W** wound around the reinforcing bar **S** is displaced from the state illustrated by the solid line in FIG. **21B** in the direction expanding in the radial direction of the loop **Ru** as indicated by the two-dot chain line. When the wire **W** wound around the reinforcing bar **S** is displaced in the direction expanding in the radial direction of the loop **Ru**, one end **WS** side of the wire **W** guided between

the fixed gripping member **70C** and the first movable gripping member **70L** by the gripping unit **70** is displaced backward. Therefore, as illustrated in FIG. **22B**, the position of the wire **W** in the radial direction of the loop **Ru** is restricted by the wall surface **54a** of the fixed guide unit **54**, whereby the displacement of the wire **W** guided to the gripping unit **70** in the radial direction of the loop **Ru** is suppressed, and occurrence of gripping failure is suppressed. In the present embodiment, even when the one end **WS** side of the wire **W** guided between the fixed gripping member **70C** and the first movable gripping member **70L** is not displaced, and the wire **W** is displaced in a direction of spreading in the radial direction of the loop **Ru**, the displacement of the wire **W** in the radial direction of the loop **Ru** is suppressed by the fixed guide unit **54**, thereby suppressing the occurrence of gripping failure.

(196) As a result, the wire **W** is wound in a loop shape around the reinforcing bar **S**. At this time, as illustrated in FIG. **20B**, the two wires **W** wound around the reinforcing bar **S** are held in a state in which they are arranged in parallel with each other without being twisted. When detecting that the movable guide unit **55** of the second guide unit **51** is opened by the output of the guide opening/closing sensor **56**, the control unit **14A** does not drive the feed motor **33a** even when the trigger **12A** is operated. Instead, notification is performed by a notifying unit (not illustrated) such as a lamp or a buzzer. This prevents occurrence of guidance failure of the wire **W**.

(197) FIG. **14** illustrates a state where the wire **W** is gripped by the gripping unit **70**. After stopping the feeding of the wire **W**, the motor **80** is driven in the normal rotation direction, whereby the motor **80** moves the movable member **83** in the direction of the arrow **F** which is the forward direction. That is, in the movable member **83**, the rotation operation interlocked with the rotation of the motor **80** is restricted by the rotation restricting member **84**, and the rotation of the motor **80** is converted into a linear movement. As a result, the movable member **83** moves in the forward direction. In conjunction with the operation of the movable member **83** moving in the forward direction, the first movable gripping member **70L** is displaced in a direction approaching the fixed gripping member **70C**, and one end **WS** side of the wire **W** is gripped.

(198) Further, the operation of the movable member **83** moving in the forward direction is transmitted to the retreat mechanism **53a**, and the guide pin **53** is retreated from the path through which the wire **W** moves.

(199) FIG. **15** illustrates a state where the wire **W** is wound around the reinforcing bar **S**. After the one end **WS** side of the wire **W** is gripped between the first movable gripping member **70L** and the fixed gripping member **70C**, and the feed motor **33a** is driven in the reverse rotation direction, the first feed gear **30L** rotates reversely and the second feed gear **30R** rotates reversely following the first feed gear **30L**.

(200) Therefore, the two wires **W** are pulled back toward the magazine **2A** and are fed in the opposite (backward) direction. In the operation of feeding the wire **W** in the backward direction, the wire **W** is wound so as to be in close contact with the reinforcing bar **S**. In this example, as illustrated in FIG. **20C**, since two wires are arranged in parallel with each other, an increase in feed resistance due to twisting of the wires **W** in the operation of feeding the wire **W** in the opposite direction is suppressed. Further, in the case where the same binding strength is to be obtained between the case where the reinforcing bar **S** is bound with a single wire as in the conventional case and the case where the reinforcing bar **S** is bound with the two wires **W** as in this example, the diameter of each wire **W** can be made thinner by using two wires **W**. Therefore, it is easy to bend the wire **W**, and the wire **W** can be brought into close contact with the reinforcing bar **S** with a small force. Therefore, the wire **W** can be reliably wound around the reinforcing bar **S** in close contact with a small force. In addition, by using two thin wires **W**, it is easy to make the wire **W** in a loop shape, and it is also possible to reduce the load at the time of cutting the wire **W**. Along with this, it is possible to downsize each motor of the reinforcing bar binding machine **1A**, and downsize the entire main body by downsizing the mechanical section. In addition, it is possible to reduce power consumption by reducing the size of the motor and reducing the load.

(201) FIG. 16 illustrates a state in which the wire W is cut. After winding the wire W around the reinforcing bar S, and stopping the feeding of the wire W, the motor 80 is driven in the normal rotation direction, thereby moving the movable member 83 in the forward direction. In conjunction with the operation of the movable member 83 moving in the forward direction, the second movable gripping member 70R is displaced in a direction approaching the fixed gripping member 70C, and the wire W is gripped. In addition, the operation of the movable member 83 moving in the forward direction is transmitted to the cutting unit 6A by the transmission mechanism 62, and the other end WE side of the wire W gripped by the second movable gripping member 70R and the fixed gripping member 70C is cut by the operation of the rotary blade portion 61.

(202) FIG. 17 illustrates a state in which the end of the wire W is bent toward the reinforcing bar S side. By moving the movable member 83 further in the forward direction after cutting the wire W, the bending portion 71 moves in the forward direction integrally with the movable member 83.

(203) As illustrated in FIGS. 22B and 22C, the bending portion 71 moves in a direction approaching the reinforcing bar S which is a forward direction indicated by an arrow F, so that the bending portion includes a bending portion 71a which is brought into contact with one end WS side of the wire W gripped by the fixed gripping member 70C and the first movable gripping member 70L. Further, the bending portion 71 moves in the direction approaching the reinforcing bar S which is the forward direction indicated by the arrow F, so that the bending portion includes a bending portion 71b which is brought in contact with the other end WE side of the wire W gripped by the fixed gripping member 70C and the second movable gripping member 70R.

(204) By moving the bending portion 71 by a predetermined distance in the forward direction indicated by the arrow F, one end WS side of the wire W gripped by the fixed gripping member 70C and the first movable gripping member 70L is pressed by the bending portion 71a to the reinforcing bar S side and is bent toward the reinforcing bar S side with the gripping position as a fulcrum.

(205) As illustrated in FIGS. 22A and 22B, the gripping unit 70 includes a slip preventing portion 75 (the protrusion portion 70Lb may also serve as the slip preventing portion 75) protruding toward the fixed gripping member 70C on the distal end side of the first movable gripping member 70L. One end WS side of the wire W gripped by the fixed gripping member 70C and the first movable gripping member 70L is bent toward the reinforcing bar S side with the slip preventing portion 75 as a fulcrum at the gripping position by the fixed gripping member 70C and the first movable gripping member 70L by moving the bending portion 71 in the forward direction indicated by the arrow F. In FIG. 22B, the second movable gripping member 70R is not illustrated.

(206) Further, by moving the bending portion 71 by a predetermined distance in the forward direction indicated by the arrow F, the other end WE side of the wire W gripped by the fixed gripping member 70C and the second movable gripping member 70R is pressed to the reinforcing bar S side by the bending portion 71b and is bent toward the reinforcing bar S side with the gripping position as a fulcrum.

(207) As illustrated in FIGS. 22A and 22C, the gripping unit 70 is provided with a slip preventing portion 76 protruding toward the fixed gripping member 70C at the distal end side of the second movable gripping member 70R. The bending portion 71 is moved in the forward direction indicated by the arrow F, so that the other end WE side of the wire W gripped by the fixed gripping member 70C and the second movable gripping member 70R is bent toward the reinforcing bar S side at the gripping position by the fixed gripping member 70C and the second movable gripping member 70R with the slip preventing portion 76 as a fulcrum. In FIG. 22C, the first movable gripping member 70L is not illustrated.

(208) FIG. 18 illustrates a state in which the wire W is twisted. After the end of the wire W is bent toward the reinforcing bar S side, the motor 80 is further driven in the normal rotation direction, whereby the motor 80 further moves the movable member 83 in the direction of the arrow F which is the forward direction. When the movable member 83 moves to a predetermined position in the

direction of the arrow F, the movable member **83** comes off from the locking of the rotation restricting member **84**, and the regulation of rotation by the rotation restricting member **84** of the movable member **83** is released. As a result, the motor **80** is further driven in the normal rotation direction, whereby the gripping unit **70** gripping the wire W rotates and twists the wire W. The gripping unit **70** is biased backward by a spring (not illustrated), and twists the wire W while applying tension thereon. Therefore, the wire W is not loosened, and the reinforcing bar S is bound with the wire W.

(209) FIG. **19** illustrates a state where the twisted wire W is released. After the wire W is twisted, the motor **80** is driven in the reverse rotation direction, so that the motor **80** moves the movable member **83** in the backward direction indicated by the arrow R. That is, in the movable member **83**, the rotation operation interlocked with the rotation of the motor **80** is restricted by the rotation restricting member **84**, and the rotation of the motor **80** is converted into a linear movement. As a result, the movable member **83** moves in the backward direction. In conjunction with the operation of the movable member **83** moving in the backward direction, the first movable gripping member **70L** and the second movable gripping member **70R** are displaced in a direction away from the fixed gripping member **70C**, and the gripping unit **70** releases the wire W. When the binding of the reinforcing bar S is completed and the reinforcing bar S is pulled out from the reinforcing bar binding machine **1A**, conventionally, the reinforcing bar S may be caught by the guide unit and it may be difficult to remove, which deteriorates workability in some cases. On the other hand, by configuring the movable guide unit **55** of the second guide unit **51** to be rotatable in the arrow H direction, when the reinforcing bar S is pulled out from the reinforcing bar binding machine **1A**, the movable guide unit **55** of the second guide unit **51** does not catch the reinforcing bar S, and thus workability is improved.

(210) <Example of Action and Effect of Reinforcing Bar Binding Machine of this Embodiment>

(211) In the reinforcing bar binding machine **1A** of the present embodiment, as illustrated in FIG. **2**, the displacement unit **34** includes a second displacement member **36** behind the first feed gear **30L** and the second feed gear **30R**, that is, between the first feed gear **30L** and the second feed gear **30R** and the handle portion **11A** in a direction substantially orthogonal to the feeding direction of the wire W. An operation button **38** for displacing the second displacement member **36**, a release lever **39** for locking and unlocking the operation button **38** are provided between the first feed gear **30L** and the second feed gear **30R** and the handle portion **11A**.

(212) In this way, by providing the mechanism for displacing the second feed gear **30R** between the second feed gear **30R** and the handle portion **11A** behind the second feed gear **30R**, there is no need to provide a mechanism for displacing the second feed gear **30R** in the feed path of the wire W below the first feed gear **30L** and the second feed gear **30R**.

(213) This makes it possible to dispose the magazine **2A** close to the wire feeding unit **3A** as compared with a configuration in which a mechanism for displacing a pair of feed gears is provided between the wire feeding unit and the magazine, thereby reducing the size of the device. Further, since the operation button **38** is not provided between the magazine **2A** and the wire feeding unit **3A**, the magazine **2A** can be disposed close to the wire feeding unit **3A**.

(214) Furthermore, since the magazine **2A** can be disposed close to the wire feeding unit **3A**, as illustrated in FIG. **11**, in the magazine **2A** housing the cylindrical reel **20**, a protrusion portion **21** which protrudes in accordance with the shape of the reel **20** can be disposed above the mounting position of the battery **15A**. Therefore, the protrusion portion **21** can be disposed close to the handle portion **11A**, and the size of the device can be reduced.

(215) In addition, since a mechanism for displacing the second feed gear **30R** is not provided in the feed path of the wire W below the first feed gear **30L** and the second feed gear **30R**, a wire loading space **22** for the wire feeding unit **3A** is formed in the magazine **2A**, and there is no constituent element which obstructs loading of the wire W, whereby loading of the wire W can be carried out easily.

(216) In the wire feeding unit configured by a pair of feed gears, a displacement member for separating one feed gear from the other feed gear, and a holding member that holds the displacement member in a state in which one feed gear is separated from the other feed gear. In such a configuration, when one feed gear is pushed in a direction away from the other feed gear due to deformation of the wire W or the like, there is a possibility that the displacement member may be locked to the holding member so that one feed gear is held in a state separated from the other feed gear.

(217) If one feed gear is held in a state separated from the other feed gear, the wire W cannot be held by the pair of feed gears, and the wire W cannot be fed.

(218) On the other hand, in the reinforcing bar binding machine **1A** of the present embodiment, as illustrated in FIG. **4**, the first displacement member **35** and the second displacement member **36** which are a displacement member for separating the second feed gear **30R** from the first feed gear **30L** and the operation button **38** and the release lever **39** for releasing the locking and the unlocking in the state where the second feed gear **30R** is separated from the first feed gear **30L** are made independent components.

(219) Accordingly, as illustrated in FIG. **4D**, when the second feed gear **30R** is pushed in a direction away from the first feed gear **30L** due to deformation of the wire W or the like, the second displacement member **36** presses the spring **37** to be displaced, but it is not locked. Therefore, the second feed gear **30R** can always be pressed in the direction of the first feed gear **30L** by the force of the spring **37**, and even if the second feed gear **30R** is temporarily separated from the first feed gear **30L**, the state in which the wire W is held by the first feed gear **30L** and the second feed gear **30R** can be restored, and the feeding of the wire W can be continued.

(220) FIGS. **23A** and **23B** are examples of the operational effects of the reinforcing bar binding machine of the present embodiment. Hereinbelow, examples of the operational effects of the reinforcing bar binding machine of this embodiment with respect to the operation of inserting the reinforcing bars into the curl guide unit and the operation of pulling the reinforcing bar from the curl guide unit will be described. For example, in the case of binding the reinforcing bars S constituting the base with the wire W, in the work using the reinforcing bar binding machine **1A**, the opening between the first guide unit **50** and the second guide unit **51** of the curl guide unit **5A** faces downward.

(221) When performing a binding operation, the opening between the first guide unit **50** and the second guide unit **51** is directed downward, and the reinforcing bar binding machine **1A** is moved downward as indicated by an arrow **Z1** as illustrated in FIG. **23A**, the reinforcing bar S enters the opening between the first guide unit **50** and the second guide unit **51**.

(222) When the binding operation is completed and the reinforcing bar binding machine **1A** is moved in the lateral direction indicated by the arrow **Z2** as illustrated in FIG. **22B**, the second guide unit **51** is pressed against the reinforcing bar S bound by the wire W, and the movable guide unit **55** on the distal end side of the second guide unit **51** rotates in the direction of the arrow H around the shaft **55b** as a fulcrum.

(223) Therefore, every time the wire W is bound to the reinforcing bar S, the binding work can be performed successively only by moving the reinforcing bar binding machine **1A** in the lateral direction without lifting the reinforcing bar binding machine **1A** every time. Therefore, (since it is sufficient to simply move the reinforcing bar binding machine **1A** in the lateral direction as compared with moving the reinforcing bar binding machine **1A** once upward and moving it downward) it is possible to reduce restrictions on the moving direction and the movement amount of the reinforcing bar binding machine **1A** in the operation of pulling out the reinforcing bar S bound to the wire W, thereby improving working efficiency.

(224) In addition, as illustrated in FIG. **21B**, the fixed guide unit **54** of the second guide unit **51** is fixed without being displaced and capable of restricting the position in the radial direction of the wire W in the binding operation described above. Accordingly, in the operation of winding the wire

W around the reinforcing bar S, the position in the radial direction of the wire W can be restricted by the wall surface **54a** of the fixed guide unit **54**, and the displacement in the direction of the wire W guided to the gripping unit **70** can be suppressed, thereby suppressing occurrence of gripping failure.

(225) FIG. **24A** illustrates an example of the operational effect of the reinforcing bar binding machine of this embodiment, and FIG. **24B** illustrates an example of an operation and a problem of the conventional reinforcing bar binding machine. Hereinbelow, an example of the operational effect of the reinforcing bar binding machine of the present embodiment as compared with the conventional one on the form of the wire W binding the reinforcing bar S will be described.

(226) As illustrated in FIG. **24B**, one end WS and the other end WE of the wire W are oriented in the opposite direction to the reinforcing bar S in the wire W bound to the reinforcing bar S in the conventional reinforcing bar binding machine. Therefore, one end WS and the other end WE of the wire W, which are the distal end side of the twisted portion of the wire W binding the reinforcing bar S largely protrude from the reinforcing bar S. If the distal end side of the wire W protrudes largely, there is a possibility that the protruding portion interferes with the operation and hinders work.

(227) Also, after the reinforcing bars S are bound, the concrete **200** is poured into the place where the reinforcing bars S are laid. At this time, in order to prevent the one end WS and the other end WE of the wire W from protruding from the concrete **200**, the thickness from the distal end of the wire W bound to the reinforcing bar S, in the example of FIG. **24B**, the thickness from the one end WS of the wire W to the surface **201** of the concrete **200** that has been poured is necessarily kept at a predetermined dimension S1. Therefore, in a configuration in which the one end WS and the other end WE of the wire W face the direction opposite to the reinforcing bar S, the thickness S12 from the laying position of the reinforcing bar S to the surface **201** of the concrete **200** becomes thick.

(228) On the other hand, in the reinforcing bar binding machine **1A** of the present embodiment, the wire W is bent by the bending portion **71** such that one end WS of the wire W wound around the reinforcing bar S is located closer to the reinforcing bar S than the first bent portion WS1 which is a bent portion of the wire W, and the other end WE of the wire W wound around the reinforcing bar S is located closer to the reinforcing bar S than the second bent portion WE1 which is a bent portion of the wire W. In the reinforcing bar binding machine **1A** of the present embodiment, the wire W is bent by the bending portion **71** such that one of the bent portion bent by the preliminary bending portion **72** in the operation of gripping the wire W by the first movable gripping member **70L** and the fixed gripping member **70C** and the bent portion bent by the fixed gripping member **70C** and the second movable gripping member **70R** in the operation of binding the wire W around the reinforcing bar S becomes the top portion which is most protruding portion in the direction away from the reinforcing bar S of the wire W.

(229) As a result, as illustrated in FIG. **24A**, the wire W bound to the reinforcing bar S in the reinforcing bar binding machine **1A** according to the present embodiment has the first bent portion WS1 between the twisted portion WT and one end WS, and one end WS side of the wire W is bent toward the reinforcing bar S side so that one end WS of the wire W is located closer to the reinforcing bar S than the first bent portion WS1. The second bent portion WE1 is formed between the twisted portion WT and the other end WE of the wire W. The other end WE side of the wire W is bent toward the reinforcing bar S side so that the other end WE of the wire W is located closer to the reinforcing bar S side than the second bent portion WE1.

(230) In the example illustrated in FIG. **24A**, two bent portions, in this example, the first bent portion WS1 and the second bent portion WE1, are formed on the wire W. Of the two, in the wire W bound to the reinforcing bar S, the first bent portion WS1 protruding most in the direction away from the reinforcing bar S (the direction opposite to the reinforcing bar S) is the top portion Wp. Both of the one end WS and the other end WE of the wire W are bent so as not to protrude beyond

the top portion Wp in the direction opposite to the reinforcing bar S.

(231) In this manner, by setting one end WS and the other end WE of the wire W so as not to protrude beyond the top portion Wp constituted by the bent portion of the wire W in the direction opposite to the reinforcing bar S, it is possible to suppress a decrease in workability due to the protrusion of the end of the wire W. Since one end WS side of the wire W is bent toward the reinforcing bar S and the other end WE side of the wire W is bent toward the reinforcing bar S side, the amount of protrusion on the distal end side from the twisted portion WT of the wire W is less than the conventional case. Therefore, the thickness S2 from the laying position of the reinforcing bar S to the surface 201 of the concrete 200 can be made thinner than the conventional one.

Therefore, it is possible to reduce the amount of concrete to be used.

(232) In the reinforcing bar binding machine 1A of the present embodiment, the wire W is wound around the reinforcing bar S by feeding in the forward direction, and one end WS side of the wire W wound around the reinforcing bar S by feeding the wire W in the opposite direction is bent toward the reinforcing bar S side by the bending portion 71 in a state of being gripped by the fixed gripping member 70C and the first movable gripping member 70L. Further, the other end WE side of the wire W cut by the cutting unit 6A is bent toward the reinforcing bar S side by the bending portion 71 in a state of being gripped by the fixed gripping member 70C and the second movable gripping member 70R.

(233) As a result, as illustrated in FIG. 22B, the gripping position by the fixed gripping member 70C and the first movable gripping member 70L is taken as a fulcrum 71c1, and as illustrated in FIG. 22C, the gripping position by the fixed gripping member 70C and the second movable gripping member 70R is taken as a fulcrum 71c2, the wire W can be bent. In addition, the bending portion 71 can apply a force that presses the wire W in the direction of the reinforcing bar S by displacement in a direction approaching the reinforcing bar S.

(234) As described above, in the reinforcing bar binding machine 1A of the present embodiment, since the wire W is gripped securely at the gripping position and the wire W is bent with the fulcrums 71c1 and 71c2, it is possible that the force pressing the wire W is reliably applied to a desired direction (the reinforcing bar S side) without being dispersed to the other direction, thereby reliably bending the ends WS and WE sides of the wire W the desired direction (the reinforcing bar S side).

(235) On the other hand, for example, in the conventional binding machine that applies a force in a direction in which the wire W is twisted in a state where the wire W is not gripped, the end of the wire W can be bent in a direction that twists the wire W, but a force to bend the wire W is applied in the state where the wire W is not gripped, so that the direction of bending the wire W is not fixed and the end of the wire W may face outward opposite to the reinforcing bar S in some cases.

(236) However, in the present embodiment, as described above, since the wire W is firmly gripped at the gripping position and the wire W is bent with the fulcrums 71c1 and 71c2, the ends WS and WE sides of the wire W can reliably be directed to the reinforcing bar S side.

(237) Further, if the end of the wire W is to be bent toward the reinforcing bar S side after twisting the wire W to bind the reinforcing bar S, there is a possibility that the binding position where the wire W is twisted is loosened and the binding strength decreases. Furthermore, when twisting the wire W to bind the reinforcing bar S and then trying to bend the wire end by applying a force in a direction in which the wire W is twisted further, there is a possibility that the binding place where the wire W is twisted is damaged.

(238) On the other hand, in the present embodiment, one end WS side and the other end WE side of the wire W are bent toward the reinforcing bar S side before twisting the wire W to bind the reinforcing bar S, so that the binding place where the wire W is twisted does not become loosened and the binding strength does not decrease. Also, after twisting the wire W to bind the reinforcing bar S, no force is applied in the direction of twisting the wire W, so that the binding place where the wire W is twisted is not damaged.

(239) FIGS. 25A and 26A show examples of operational effects of the reinforcing bar binding machine according to the present embodiment, and FIGS. 25B and 26B show examples of the operations and problems of the conventional reinforcing bar binding machine. Hereinbelow, an example of the operational effect of the reinforcing bar binding machine according to the present embodiment as compared with the conventional one will be described in terms of prevention of the wire W coming out from the gripping unit in the operation of winding the wire W around the reinforcing bar S.

(240) As illustrated in FIG. 25B, the conventional gripping unit 700 of the reinforcing bar binding machine includes a fixed gripping member 700C, a first movable gripping member 700L, and a second movable gripping member 700R, and a length restricting unit 701 against which the wire W wound around the reinforcing bar S abuts is provided in the first movable gripping member 700L.

(241) In the operation of feeding the wire W in the backward direction (pulling back) and winding it around the reinforcing bar S and the operation of twisting the wire W by the gripping unit 700, the wire W gripped by the fixed gripping member 700C and the first movable gripping member 700L is likely to come off when the distance N2 from the gripping position of the wire W by the fixed gripping member 700C and the first movable gripping member 700L to the length restricting unit 701 is short.

(242) In order to make it difficult for the gripped wire W to come off it is simply necessary to lengthen the distance N2. However, for this purpose, it is necessary to lengthen the distance from the gripping position of the wire W in the first movable gripping member 700L to the length restricting unit 701.

(243) However, if the distance from the gripping position of the wire W in the first movable gripping member 700L to the length restricting unit 701 is increased, the size of the first movable gripping member 700L is increased. Therefore, in the conventional configuration, it is not possible to lengthen the distance N2 from the gripping position of the wire W by the fixed gripping member 700C and the first movable gripping member 700L to one end WS of the wire W.

(244) On the other hand, as illustrated in FIG. 27A, in the gripping unit 70 of the present embodiment, the length restricting unit 74 where the wire W abuts is set to be a separate component independent from the first movable gripping member 70L.

(245) This makes it possible to lengthen the distance N1 from the gripping position of the wire W in the first movable gripping member 70L to the length restricting unit 74 without increasing the size of the first movable gripping member 70L.

(246) Therefore, even if the first movable gripping member 70L is not enlarged, it is possible to prevent the wire W gripped by the fixed gripping member 70C and the first movable gripping member 70L from coming off during the operation of feeding the wire W in the backward direction to wind around the reinforcing bar S and the operation of twisting the wire W by the gripping unit 70.

(247) As illustrated in FIG. 26B, the conventional gripping unit 700 of the reinforcing bar binding machine is provided with, on the surface of the first movable gripping member 700L facing the fixed gripping member 700C, a protrusion protruding toward the fixed gripping member 700C and a recess into which the fixed gripping member 700C is inserted, thereby forming a preliminary bending portion 702.

(248) As a result, in the operation of gripping the wire W by the first movable gripping member 700L and the fixed gripping member 700C, one end WS side of the wire W protruding from the gripping position by the first movable gripping member 700L and the fixed gripping member 700C is bent, and in the operation of feeding the wire W in the backward direction to wind around the reinforcing bar S and the operation of twisting the wire W by the gripping unit 700, the effect of preventing the wire W from coming off can be obtained.

(249) However, since one end WS side of the wire W is bent inward toward the wire W passing between the fixed gripping member 700C and the second movable gripping member 700R, the bent

one end WS side of the wire W may be caught in contact with the wire W to be fed in the backward direction for winding around the reinforcing bar S.

(250) When the bent one end WS side of the wire W is caught by the wire W that is fed in the backward direction for winding around the reinforcing bar S, there is a possibility that the winding of the wire W becomes insufficient or the twisting of the wire W is insufficient.

(251) On the other hand, in the gripping unit **70** of the present embodiment, as illustrated in FIG. **28A**, on the surface facing the first movable gripping member **70L** of the fixed gripping member **70C**, a protrusion protruding toward the first movable gripping member **70L** and a recess into which the first movable gripping member **70L** is inserted are provided to form the preliminary bending portion **72**.

(252) Therefore, in the operation of gripping the wire W by the first movable gripping member **70L** and the fixed gripping member **70C**, one end WS side of the wire W protruding from the gripping position by the first movable gripping member **70L** and the fixed gripping member **70C** is bent, and in the operation of feeding the wire W in the backward direction to wind around the reinforcing bar S, and the operation of twisting the wire W by the gripping unit **70**, the effect of preventing the wire W from coming off can be obtained.

(253) One end WS side of the wire W is bent to the outside opposite to the wire W passing between the fixed gripping member **70C** and the second movable gripping member **70R**, so that it is suppressed that the bent one end WS side of the wire W is in contact with the wire W fed in the backward direction to wind around the reinforcing bar S.

(254) Thus, in the operation of feeding the wire W in the backward direction to wind around the reinforcing bar S, it is prevented that the wire W comes off from the gripping unit **70**, thereby surely winding the wire W, and in the operation of twisting the wire W, it is possible to reliably perform the binding of the wire W.

(255) FIGS. **27A**, **27B**, and **28A** show examples of operational effects of the reinforcing bar binding machine of the present embodiment, and FIGS. **27C**, **27D**, and **28B** are examples of the operation and problems of the conventional reinforcing bar binding machine. Hereinbelow, an example of the operational effects of the reinforcing bar binding machine according to the present embodiment as compared with the related art will be described with respect to the operation of binding the reinforcing bar S with the wire W.

(256) As illustrated in FIG. **27C**, in the conventional configuration in which one wire Wb having a predetermined diameter (for example, about 1.6 mm to 2.5 mm) is wound around the reinforcing bar S, as illustrated in FIG. **27D**, since the rigidity of the wire Wb is high, unless the wire Wb is wound around the reinforcing bar S with a sufficiently large force, slack J occurs during the operation of winding the wire Wb, and a gap is generated between the wire and the reinforcing bar S.

(257) On the other hand, as illustrated in FIG. **27A**, in the present embodiment in which two wires W having a small diameter (for example, about 0.5 mm to 1.5 mm) are wound around the reinforcing bar S as compared with the conventional case, as illustrated in FIG. **27B**, since the rigidity of the wire W is lower than that of the conventional wire, even if the wire W is wound around the reinforcing bar S with a lower force than the conventional case, slack in the wire W occurring during the operation of winding the wire W is suppressed, and the wire is surely wound around the reinforcing bar S at the linear portion K. Considering the function of binding the reinforcing bar S with the wire W, the rigidity of the wire W varies not only by the diameter of the wire W but also by the material thereof etc. For example, in the present embodiment, the wire W having a diameter of about 0.5 mm to 1.5 mm is described as an example. However, if the material of the wire W is also taken into consideration, between the lower limit value and the upper limit value of the diameter of the wire W, at least a difference of about tolerance may occur.

(258) Further, as illustrated in FIG. **28B**, in the conventional configuration in which one wire Wb having a predetermined diameter is wound around the reinforcing bar S and twisted, since the

rigidity of the wire Wb is high, even in the operation of twisting the wire Wb, the slack of the wire Wb is not eliminated, and a gap L is generated between the wire and the reinforcing bar S.

(259) On the other hand, as illustrated in FIG. 28A, in the present embodiment in which two wires W having a smaller diameter are wound around the reinforcing bar S and twisted as compared with the related art, the rigidity of the wire W is lower as compared with the conventional one, by the operation of twisting the wire W, the gap M between the reinforcing bar S and the wire can be suppressed small as compared with the conventional case, whereby the binding strength of the wire W is improved.

(260) By using the two wires W, it is possible to equalize the reinforcing bar holding force as compared with the conventional case, and to suppress the deviation between the reinforcing bars S after the binding. In the present embodiment, two wires W are simultaneously (together) fed, and the reinforcing bars S are bound using the two wires W fed simultaneously (together). Feeding the two wires W at the same time means that when one wire W and the other wire W are fed at substantially the same speed, that is, when the relative speed of the other wire W to one wire W is substantially 0. In this example, the meaning is not necessarily limited to this meaning. For example, even when one wire W and the other wire W are fed at different speeds (timings), the two wires W are arranged in parallel with each other and advance in parallel in the feed path of the wire W, so, as long as the wire W is set to be wound around the reinforcing bar S in the parallel state, it means that two wires are fed at the same time. In other words, the total area of the cross-sectional area of each of the two wires W is a factor determining the reinforcing bar holding force, so even if the timings of feeding the two wires W are deviated, in terms of securing the reinforcing bar holding force, the same result can be obtained. However, compared to the operation of shifting the timing of feeding the two wires W, since it is possible to shorten the time required for feeding for the operation of simultaneously (together) feeding the two wires W, it is preferable to feed the two wires W simultaneously (together), resulting in improvement of the binding speed.

(261) <Modified Example of Reinforcing Bar Binding Machine According to this Embodiment>

(262) FIGS. 29A and 29B are configuration diagrams illustrating modified examples of the second guide unit of the present embodiment. The displacement direction of the movable guide unit 55 of the second guide unit 51 is restricted by the guide shaft 55c and the guide groove 55d along the displacement direction of the movable guide unit 55. For example, as illustrated in FIG. 29A, the movable guide unit 55 includes the guide groove 55d extending along the direction in which the movable guide unit 55 moves with respect to the first guide unit 50, that is, the direction in which the movable guide unit 55 moves closer to and away from the first guide unit 50. The fixed guide unit 54 includes the guide shaft 55c which is inserted into the guide groove 55d and is movable in the guide groove 55d. Consequently, the movable guide unit 55 is displaced from the guide position to the retreat position by the parallel movement in the direction in which the movable guide unit 55 comes into contact with and separates from the first guide unit 50 (up and down direction in FIG. 29A).

(263) Further, as illustrated in FIG. 29B, a guide groove 55d extending in the forward and backward direction may be provided in the movable guide unit 55. As a result, the movable guide unit 55 is displaced from the guide position to the retreat position by movement in the forward and backward direction in which protruding from the front end, which is one end of the main body 10A, and retreating to the inside of the main body 10A are performed. The guide position in this case is a position where the movable guide unit 55 protrudes from the front end of the main body 10A so that the wall surface 55a of the movable guide unit 55 exists at a position where the wire W forming the loop Ru passes. The retreat position is a state in which all or a part of the movable guide unit 55 has entered the inside of the main body 10A. Further, a configuration may be adopted in which the movable guide unit 55 is provided with a guide groove 55d extending in an oblique direction along the direction of contacting and separating from the first guide unit 50 and in the forward and backward direction. The guide groove 55d may be formed in a straight line shape or a

curved line shape such as a circular arc.

(264) As another modified example of the reinforcing bar binding machine **1A** of the present embodiment, the configuration is described in which two wires **W** are used, but the reinforcing bar **S** may be bound with one wire **W** or two or more wires **W**. In the reinforcing bar binding machine **1A** according to the present embodiment, the length restricting portion **74** is provided in the first guide unit **50** of the curl guide unit **5A**, but may be provided in the first movable gripping member **70L** or the like, or another location, as long as it is a component independent of the gripping unit **70**, for example, a structure that supports gripping unit the gripping unit **70**.

(265) Further, before the operation of bending the one end **WS** side and the other end **WE** side of the wire **W** toward the reinforcing bar **S** side by the bending portion **71** is completed, the rotation operation of the gripping unit **70** is started, and thus the operation of twisting the wire **W** may be started. Further, after starting the operation of twisting the wire **W** by starting the rotation operation of the gripping unit **70**, before the operation of twisting the wire **W** is completed, the operation of bending the one end **WS** side and the other end **WE** side toward the reinforcing bar **S** side by the bending portion **71** may be started and ended.

(266) In addition, although the bending portion **71** is formed integrally with the movable member **83** as a bending means, the gripping unit **70** and the bending portion **71** may be driven by an independent driving unit such as a motor. Further, instead of the bending portion **71**, as a bending unit, a bending portion formed in a concave-convex shape, or the like may be provided in any of the fixed gripping member **70C**, the first movable gripping member **70L**, and the second movable gripping member **70R** to apply a bending force by which the wire **W** is bent toward the reinforcing bar **S** in the operation of gripping the wire **W**.

(267) FIGS. **30A**, **30B**, **30C**, **30D**, and **30E** are views illustrating modified examples of the parallel guide of the present embodiment. In the parallel guide **4B** illustrated in FIG. **30A**, the cross-sectional shape of the opening **4BW**, that is, the cross-sectional shape of the opening **4BW** in a direction orthogonal to the feeding direction of the wire **W** is formed in a rectangular shape, and the longitudinal direction and the lateral direction of the opening **4BW** are formed in a straight shape. In the parallel guide **4B**, the length **L1** in the longitudinal direction of the opening **4BW** is slightly longer than the diameter **r** of a plurality of the wires **W** in a form in which the wires **W** are arranged in parallel along the radial direction, and the length **L2** in the lateral direction is slightly longer than the diameter **r** of one wire **W**. In the parallel guide **4B** in this example, the length **L1** of the opening **4BW** in the longitudinal direction is slightly longer than the diameter **r** of two wires **W**.

(268) In the parallel guide **4C** illustrated in FIG. **30B**, the longitudinal direction of the opening **4CW** is formed in a straight shape and the lateral direction is formed in a triangular shape. In the parallel guide **4C**, in order that a plurality of wires **W** are arranged in parallel in the longitudinal direction of the opening **4CW** and the wire **W** can be guided by an inclined plane in the lateral direction, the longitudinal length **L1** of the opening **4CW** is slightly longer than the diameter **r** of the plurality of the wires **W** in the form in which the wires **W** are arranged along the radial direction, and the lateral length **L2** is slightly longer than the diameter **r** of one wire **W**.

(269) In the parallel guide **4D** illustrated in FIG. **30C**, the longitudinal direction of the opening **4DW** is formed in a curved shape which is curved inward in a convex shape and the lateral direction is formed in a circular arc shape. That is, the opening shape of the opening **4DW** is formed in a shape that conforms to the outer shape of the parallel wires **W**. In the parallel guide **4D**, the length **L1** in the longitudinal direction of the opening **4DW** is slightly longer than the diameter **r** of the plurality of the wires **W** in the form in which the wires **W** are arranged along the radial direction, the length **L2** in the lateral direction is slightly longer than the diameter **r** of one wire **W**. In the parallel guide **4D**, in the present example, the length **L1** in the longitudinal direction has a length slightly longer than the diameter **r** of two wires **W**.

(270) In the parallel guide **4E** illustrated in FIG. **30D**, the longitudinal direction of the opening **4EW** is formed in a curved shape curved outward in a convex shape, and the lateral direction is

formed in a circular arc shape. That is, the opening shape of the opening **4EW** is formed in an elliptical shape. The parallel guide **4E** has a length **L1** in the longitudinal direction of the opening **4EW** which is slightly longer than the diameter **r** of the plurality of the wires **W** in the form in which the wires **W** are arranged along the radial direction, and a length **L2** in the lateral direction is slightly longer than the diameter **r** of one wire **W**. In this example, the parallel guide **4E** has a length **L1** in the longitudinal direction slightly longer than the diameter **r** of two wires **W**.

(271) The parallel guide **4F** illustrated in FIG. **30E** includes a plurality of openings **4FW** matching the number of wires **W**. Each wire **W** is passed through another opening **4FW** one by one. In the parallel guide **4F**, each opening **4FW** has a diameter (length) **L1** slightly longer than the diameter **r** of the wire **W**, and by the direction in which the openings **4FW** are arranged, the direction in which a plurality of wires **W** are arranged in parallel is restricted.

(272) FIG. **31** is a configuration diagram illustrating a modified example of the guide groove of this embodiment. The guide groove **52B** has a width (length) **L1** and a depth **L2** slightly longer than the diameter **r** of the wire **W**. Between one guide groove **52B** through which one wire **W** passes and the other guide groove **52B** through which the other wire **W** passes, a section wall portion is formed along the feeding direction of the wire **W**. The first guide unit **50** restricts the direction in which a plurality of wires are arranged in parallel with each other by the direction in which the plurality of guide grooves **52B** are arranged.

(273) FIGS. **32** to **34** are views illustrating an example of a displacement unit of another embodiment, and FIG. **35** is an external view illustrating an example of a reinforcing bar binding machine of another embodiment. A displacement unit **340** is an example of a displacement unit, and includes a first displacement member **350** that is displaced in directions indicated by arrows **V1** and **V2** by a rotation operation with a shaft **350a** as a fulcrum, and displaces a second feed gear **30R** in a direction separating from a first feed gear **30L**. Furthermore, the displacement unit **340** includes a second displacement member **360** for displacing the first displacement member **350**.

(274) The first displacement member **350** is a long plate-like member and has one end side rotatably supported to the shaft **350a** and the other end side to which the second feed gear **30R** is rotatably supported by a shaft **300R**. It is noted that the shape of the first displacement member **350** is not limited to the long plate-like member. Furthermore, the first displacement member **350** includes a pressed portion **350b** pressed from the second displacement member **360** in the range of a thickness **t** along the axial direction of the second feed gear **30R** supported via the shaft **300R**, preferably, in the vicinity of a position of a second feed groove **32R**.

(275) The pressed portion **350b** is disposed so as to extend toward a radial direction of the second feed gear **30R** from the shaft **300R**. The pressed portion **350b** has a U shape and is attached to the shaft **300R** so as to sandwich the second feed gear **30R** with the U-shaped opening. It is noted that the shape of the pressed portion **350b** is not limited to the U shape.

(276) The second displacement member **360** is rotatably supported to a shaft **360a** and is displaced in directions indicated by arrows **W1** and **W2** by a rotation operation with the shaft **360a** as a fulcrum. The second displacement member **360** includes a pressing portion **360b**, which presses the pressed portion **350b** of the first displacement member **350**, at one end side at which the shaft **360a** is sandwiched. The pressing portion **360b** presses the pressed portion **350b** in the range of the thickness **t** along the axial direction of the second feed gear **30R**, preferably, in the vicinity of the position of the second feed groove **32R**.

(277) The first displacement member **350** is displaced with a rotation operation with the shaft **350a** as a fulcrum and the second displacement member **360** is displaced with a rotation operation with the shaft **360a** as a fulcrum, but their shafts are not parallel to each other. The pressing portion **360b** is configured by a convex arc along the rotation operation with the shaft **360a** as a fulcrum. Furthermore, the pressed portion **350b** is configured by a convex arc along a rotation operation with the shaft **300R** as a fulcrum. As a result, contact points between the pressing portion **360b** and the pressed portion **350b** are suppressed from being largely deviated by the rotation operations of

the first displacement member **350** and the second displacement member **360**.

(278) Moreover, in the first displacement member **350**, at least the pressed portion **350b** or the entire is configured by iron, and in the second displacement member **360**, at least the pressing portion **360b** or the entire is configured by iron. As a result, abrasion of a contact point between the pressing portion **360b** and the pressed portion **350b** is suppressed.

(279) The second displacement member **360** includes a spring abutting portion **370a**, which is abutted by a spring **370** configured by a compression coil spring for example, at the other end side at which the shaft **360a** is sandwiched. The spring **370** is urged in a direction of pushing the spring abutting portion **370a**. Therefore, one end side of the second displacement member **360**, that is, the pressing portion **360b** enters a state of pressing the pressed portion **350b** by urging force of the spring **370**.

(280) The spring **370** presses the second displacement member **360** and the pressing portion **360b** of the second displacement member **360** presses the pressed portion **350b** of the first displacement member **350**, so that the second feed gear **30R** is pressed in the direction of the first feed gear **30L**.

(281) As a result, two wires **W** are sandwiched by a first feed groove **32L** of the first feed gear **30L** and a second feed groove **32R** of the second feed gear **30R**. Furthermore, a tooth portion **31L** of the first feed gear **30L** and a tooth portion **31R** of the second feed gear **30R** mesh with each other.

(282) The displacement unit **340** includes an operation button **380** for pressing the second displacement member **360** against the urging force of the spring **370**. Furthermore, the displacement unit **340** includes a release lever **390** for fixing the operation button **380** in a predetermined state, that is, a state in which the operation button **380** presses the second displacement member **360**, and releases the fixing.

(283) The operation button **380** is an example of an operation member, and is provided at a position facing the spring **370** via the second displacement member **360**. In the operation button **380**, an operation part **380b** protrudes outward from one side surface of a main body **10A**, and is movably supported to the main body **10A** in a pushing direction with respect to the main body **10A** indicated by an arrow **T1** and in a direction of protruding from the main body **10A** indicated by an arrow **T2**. The operation part **380b** of the operation button **380** is pushed in the direction of the arrow **T1** in which the main body **10A** is pushed, so that the operation button **380** and the second displacement member **360**, by which the spring **370** is sandwiched, are rotated in the direction of the arrow **T1**.

(284) The operation button **380** includes a locking recess **380a** to which the release lever **390** is locked at a wire loading position where the wire **W** can be loaded by separating the first feed gear **30L** and the second feed gear **30R**. The locking recess **380a** is configured by providing a recess at a front side of the operation button **380** so as to face the release lever **390** in the present example.

(285) The release lever **390** is an example of a release member, and is supported so as to be movable in directions indicated by arrows **U1** and **U2** intersecting the movement direction of the operation button **380** by a rotation operation with a shaft **390c** as a fulcrum.

(286) The release lever **390** includes a locking protrusion **390a** engaged with the locking recess **380a** formed in the operation button **380** when the operation button **380** is pressed to a predetermined state. Accordingly, when the operation button **380** is pressed to the predetermined state, the operation button **380** is fixed at the position by the release lever **390**. The release lever **390** includes an operation part **390d** for releasing the fixing. The operation part **390d** protrudes outward from one side surface of the main body **10A**. The release lever **390** operates the operation part **390d** to move in a direction of separating from the operation button **380**, so that the locking protrusion **390a** disengages from the locking recess **380a**.

(287) The release lever **390**, for example, is urged in the direction of the arrow **U1** toward the operation button **380** by a spring **390b** configured by a torsion coil spring, so that the locking protrusion **390a** abuts the operation button **380**.

(288) FIGS. **36** to **38** are explanatory views illustrating an example of an operation of the displacement unit of another embodiment and illustrate an operation of releasing pressing of the

second feed gear **30R**. When the operation button **380** is pushed in the direction of the arrow **T1**, the second displacement member **360** is rotated in the direction of the arrow **W1** with the shaft **360a** as a fulcrum while compressing the spring **370**. As a result, the pressing portion **360b** of the second displacement member **360** is separated from the pressed portion **350b** of the first displacement member **350**.

(289) When the operation button **380** is pushed in the direction of the arrow **T1** to a position at which the locking recess **380a** faces the locking protrusion **390a** of the release lever **390**, the release lever **390** is rotated by restoring force of the spring **390b** in the direction of the arrow **U1** with the shaft **390c** as a fulcrum by the spring **390b**. As a result, the locking protrusion **390a** of the release lever **390** enters the locking recess **380a** of the operation button **380**, so that the operation button **380** is held in the state of pressing the second displacement member **360**. Thus, at the time of loading of the wire **W**, it is not necessary to continuously push the operation button **380**.

(290) FIGS. **39** to **41** are explanatory views illustrating an example of an operation of the displacement unit of another embodiment and illustrate an operation of loading the wire **W** between the first feed gear **30L** and the second feed gear **30R**. In the state in which the pressing portion **360b** of the second displacement member **360** is separated from the pressed portion **350b** of the first displacement member **350**, the first displacement member **350** for supporting the second feed gear **30R** can be freely rotated with the shaft **350a** as a fulcrum.

(291) As a result, when two wires **W** arranged in parallel are inserted between the first feed gear **30L** and the second feed gear **30R**, the first displacement member **350** is rotated in the direction of the arrow **V1** with the shaft **350a** as a fulcrum, so that the second feed gear **30R** is separated from the first feed gear **30L**. Thus, the two wires **W** arranged in parallel are inserted between the first feed groove **32L** of the first feed gear **30L** and the second feed groove **32R** of the second feed gear **30R**.

(292) FIGS. **42** to **44** are explanatory views illustrating an example of an operation of the displacement unit of another embodiment and illustrate an operation of releasing holding of the operation button **380**. After the wire **W** is inserted between the first feed gear **30L** and the second feed gear **30R**, the release lever **390** is rotated in the direction of the arrow **U2** with the shaft **390c** as a fulcrum. As a result, the locking protrusion **390a** of the release lever **390** is pulled out from the locking recess **380a** of the operation button **380**.

(293) FIGS. **45** to **47** are explanatory views illustrating an example of an operation of the displacement unit of another embodiment and illustrate an operation of pressing the second feed gear **30R** to the first feed gear **30L**. When the locking protrusion **390a** of the release lever **390** is pulled out from the locking recess **380a** of the operation button **380** by operating the release lever **390**, the second displacement member **360** is rotated by restoring force of the spring **370** in the direction of the arrow **W2** with the shaft **360a** as a fulcrum.

(294) When the second displacement member **360** is rotated in the direction of the arrow **W2**, the pressing portion **360b** of the second displacement member **360** presses the pressed portion **350b** of the first displacement member **350**, so that the first displacement member **350** is rotated in the direction of the arrow **V2** with the shaft **350a** as a fulcrum and the second feed gear **30R** is pressed in the direction of the first feed gear **30L** by the force of the spring **370**.

(295) As a result, in the state in which the two wires **W** are arranged in parallel, the two wires **W** are sandwiched by the first feed groove **32L** of the first feed gear **30L** and the second feed groove **32R** of the second feed gear **30R**. The tooth portion **31L** of the first feed gear **30L** and the tooth portion **31R** of the second feed gear **30R** mesh with each other.

(296) Moreover, the second displacement member **360** is rotated in the direction of the arrow **W2**, so that the operation button **380** moves in the direction of the arrow **T2**.

(297) The pressed portion **350b** of the first displacement member **350** is pressed by the pressing portion **360b** of the second displacement member **360**, so that force for pressing the vicinity of the position of the second feed groove **32R** is transferred via the shaft **300R** and the second feed gear

30R is pressed in the direction of the first feed gear **30L**.

(298) As a result, the second feed gear **30R** is suppressed from being inclined with respect to the first feed gear **30L**, so that biased load is suppressed from being applied to the first feed gear **30L** and the second feed gear **30R**.

(299) Thus, biased abrasion of the first feed gear **30L** and the second feed gear **30R** is suppressed. Furthermore, the wire **W** is suppressed from being pulled out from the first feed groove **32L** of the first feed gear **30L** and the second feed groove **32R** of the second feed gear **30R**.

(300) FIG. **48** is an external view illustrating an example of a reinforcing bar binding machine of another embodiment. The operation part **380b** of the operation button **380** and the operation part **390d** of the release lever **390** are provided above a magazine **2A** at one side surface of the main body **10A** and a front side of a trigger **12A**. A finger abutment part **16** for abutting fingers is provided above the magazine **2A** at the other side surface of the main body **10A** and the front side of the trigger **12A**.

(301) As a result, when a handle part **11A** is held by a single hand, it is possible to operate the operation part **380b** of the operation button **380** by a single hand in the state of sandwiching the operation part **380b** of the operation button **380** and the finger abutment part **16**. Furthermore, it is possible to operate the operation part **390d** of the release lever **390** by a single hand in the state of sandwiching the operation part **390d** of the release lever **390** and the finger abutment part **16**. Thus, it is possible to operate the operation button **380** and the release lever **390** without placing a reinforcing bar binding machine **1A** at a work place and the like.

(302) It is noted that since it is sufficient if it is a mechanism which can be fixedly held and released between the operation button **380** and the release lever **390**, a mechanism of a locking member having an locking protrusion shape at the operation button **380** side and an locking recess shape at the release lever **390** side may be provided.

(303) In another modified example of the present embodiment, instead of a configuration of simultaneously feeding a plurality of wires **W**, a configuration may be adopted in which after the wires **W** are wound around a reinforcing bar **S** one by one so as to wind the plurality of wires, the plurality of wires are fed in a reverse direction and wound around the reinforcing bar **S**.

(304) It is noted that the present invention can also be applied to a binding machine that binds pipes or the like as a binding object with a wire.

(305) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2015-145285 filed on Jul. 22, 2015 and Japanese Patent Application No. 2016-136069 filed on Jul. 8, 2016, the entire contents of which are incorporated herein by reference.

REFERENCE SIGNS LIST

(306) **1A**: reinforcing bar binding machine, **10A**: main body, **11A**: handle portion, **2A**: magazine (housing unit), **20**: reel, **22**: wire loading space, **3A**: wire feeding unit (feeding unit), **4A**: parallel guide (feeding unit), **5A**: curl guide unit (feeding unit), **6A**: cutting unit, **7A**: binding portion (binding unit), **8A**: binding unit driving mechanism, **30L**: first feed gear (feeding member, rotary feeding member), **30R**: second feed gear (feeding member, rotary feeding member), **34**: displacement unit (displacement unit), **35**: first displacement member (displacement member), **36**: second displacement member (displacement member), **37**: spring, **38**: operation button (operation member), **39**: release lever (release member), **W**: wire

Claims

1. A binding machine comprising: a housing that houses a wire; a feeding unit that includes first and second feeding members between which the wire is sandwiched and fed; a curl guide that winds the wire fed from the feeding unit, around a binding object; a binding unit that twists the wire wound around the binding object; and a displacement unit that displaces the second feeding member toward and away from the first feeding member, wherein the displacement unit is provided

in a direction intersecting a feeding direction of the wire with respect to the feeding members, the displacement unit includes: a first displacement member including an end portion and a base portion; and a second displacement member configured to rotate the first displacement member by pressing the end portion of the first displacement member; an operation member which displaces the second displacement member; and a release member which fixes a position of the operation member and which releases a fixed position of the operation member.

2. The binding machine according to claim 1, further comprising a wire loading space located between the feeding unit and the housing to load the wire between the first and second feeding members.

3. The binding machine according to claim 2, wherein the wire loading space is located inside the housing.

4. The binding machine according to claim 1, further comprising: a main body; and a handle portion that protrudes from the main body in one direction, wherein the feeding unit is located between the binding unit and the housing, and the displacement unit is located between the feeding unit and the handle portion in the main body.

5. The binding machine according to claim 1, further comprising: a main body; and a handle portion that protrudes from the main body in one direction, wherein the displacement unit includes: the first displacement member which displaces the second feeding member in a direction where the second feeding member is movable toward and away from the first feeding member; an operation member which displaces the first displacement member; and a release member which fixes a position of the operation member and which releases a fixed position of the operation member, and the operation member and the release member are located between the feeding members and the handle portion in the main body.

6. The binding machine according to claim 5, wherein the end portion is supported on the second feeding member and the base portion is supported on the main body such that the second feeding member is moved toward and away from the first feeding member by a rotation operation of the second displacement member.

7. The binding machine according to claim 1, further comprising: a main body; and a handle portion that protrudes from the main body in one direction, wherein the displacement unit includes: the first displacement member which displaces the second feeding member in a direction where the second feeding member is movable toward and away from the first feeding member; an operation member which displaces the first displacement member and which protrudes outward from the main body, and the first displacement member is located between the feeding members and the handle portion in the main body.

8. The binding machine according to claim 1, wherein the release member fixes the second displacement member in a state where the first displacement member is pressed by the second displacement member, and the release member releases a fixed state of the second displacement member.

9. The binding machine according to claim 1, wherein the operation member and the release member include protrusion and recess portions as a locking mechanism provided between the operation member and the release member to fix the position of the operation member and to release a fixed position of the operation member.

10. A binding machine comprising: a housing that houses a wire; a feeding unit that includes first and second feeding members between which the wire is sandwiched and fed; a curl guide that winds the wire fed from the feeding unit, around a binding object; a binding unit that twists the wire wound around the binding object; and a displacement unit that displaces the second feeding member toward and away from the first feeding member, wherein the displacement unit is provided in a direction intersecting a feeding direction of the wire with respect to the feeding members, the first and second feeding members are a pair of rotary feeding members which feeds the wire by a rotation operation, the displacement unit includes: a first displacement member which displaces the

second feeding member in a direction where the second feeding member is movable toward and away from the first feeding member; a second displacement member which displaces the first displacement member; an operation member which displaces the second displacement member; and a release member which fixes a position of the operation member and which releases a fixed position of the operation member, the first displacement member includes a pressed portion which is pressed by the second displacement member within a range of a thickness in an axial direction of the rotary feeding members, and the second displacement member includes a pressing portion which presses the pressed portion within the range of the thickness in the axial direction of the rotary feeding members.

11. A binding machine comprising: a housing that houses a wire; first and second feeding gears between which the wire is sandwiched and fed; a curl guide that winds the wire fed from the first and second feeding gears, around a binding object; a binding unit that twists the wire wound around the binding object; and a displacement unit provided in a direction intersecting a feeding direction of the wire with respect to the first and second feeding gears, the displacement unit including: a first displacement arm including an end portion and a base portion, wherein the first displacement arm is configured to displace the second feeding gear in a direction where the second feeding gear is movable toward and away from the first feeding gear; a second displacement arm configured to rotate the first displacement arm by pressing the end portion of the first displacement arm; an operation button which displaces the second displacement arm; and a release lever which fixes a position of the operation button and which releases a fixed position of the operation button.

12. The binding machine according to claim 11, further comprising a wire loading space located between the first and second feeding gears and the housing to load the wire between the first and second feeding gears.

13. The binding machine according to claim 12, wherein the wire loading space is located inside the housing.

14. The binding machine according to claim 11, further comprising: a main body; and a handle portion that protrudes from the main body in one direction, wherein the first and second feeding gears are located between the binding unit and the housing, and the displacement unit is located between the first and second feeding gears and the handle portion in the main body.

15. The binding machine according to claim 11, further comprising: a main body; and a handle portion that protrudes from the main body in one direction, wherein: the operation button and the release lever are located between the first and second feeding gears and the handle portion in the main body.

16. The binding machine according to claim 15, wherein the end portion is supported on the second feeding gear and the base portion is supported on the main body such that the second feeding gear is moved toward and away from the first feeding gear by a rotation operation of the second displacement arm.

17. The binding machine according to claim 11, further comprising: a main body; and a handle portion that protrudes from the main body in one direction, wherein: the operation button protrudes outward from the main body, and the first displacement arm is located between the first and second feeding gears and the handle portion in the main body.

18. The binding machine according to claim 11, wherein the release lever fixes the second displacement arm in a state where the first displacement arm is pressed by the second displacement arm, and the release lever releases a fixed state of the second displacement arm.

19. The binding machine according to claim 11, wherein the operation button and the release lever include protrusion and recess portions as a locking mechanism provided between the operation button and the release lever to fix the position of the operation button and to release a fixed position of the operation button.

20. A binding machine comprising: a housing that houses a wire; first and second feeding gears between which the wire is sandwiched and fed; a curl guide that winds the wire fed from the first

and second feeding gears, around a binding object; a binding unit that twists the wire wound around the binding object; and a displacement unit provided in a direction intersecting a feeding direction of the wire with respect to the first and second feeding gears, the first and second feeding gears are configured to feed the wire by a rotation operation, the displacement unit including: a first displacement arm which displaces the second feeding gear in a direction where the second feeding gear is movable toward and away from the first feeding gear; a second displacement arm which displaces the first displacement arm; an operation button which displaces the second displacement arm; and a release lever which fixes a position of the operation button and which releases a fixed position of the operation button.
